

Going Public over the Business Cycle*

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Abstract

This paper analyzes the role of Initial Public Offerings (IPOs) on employment cyclicalities. Using Compustat data, I show that IPOs are cyclical in numbers and compositions and that the economic conditions at the time of IPO have long-lasting effects on subsequent firm growth. An important factor driving this result is the procyclical amount of capital raised from IPOs. Building on this evidence, I develop a firm dynamics model in which private firms decide whether to go public, while public and private firms decide about employment, investment, and exit in the presence of borrowing constraints. Crucially for the model mechanism, public firms face a looser borrowing constraint. The model is calibrated to match selected features of the U.S. non-financial firm sector and replicates the procyclical number of IPOs, selection patterns, and procyclical capital injections observed in the data. Using my model, I find that if IPOs were acyclical, the volatility of employment would be reduced by 10.4 percent. This amplification operates through two channels: first, a decline in the public firm share exacerbates capital misallocation; second, a lower propensity to go public reduces potential entrants' expected future value, decreasing entry rates. These findings suggest that policies mitigating IPO cyclicalities during recessions could facilitate a faster recovery in aggregate employment.

Keywords: Firm dynamics; Growth; Business Cycle; Macroeconomics; IPO

JEL codes: D25, E32, G32, H32

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1 Introduction

Labor markets tend to recover slowly after recessions, and financial frictions are often considered as an important driver of this sluggish rebound. While debt financing is a fundamental channel for firms to secure external capital, equity financing also plays a critical role. In particular, an Initial Public Offering (IPO)—a firm’s first equity issuance—may contribute to labor market recovery by promoting the growth of firms. Evidence indicates that IPOs significantly impact employment at the firm level, increasing it by an average of 37 percent in the IPO year (Borisov et al., 2021). More importantly, equity financing is uniquely susceptible to excess volatility in stock prices (Shiller, 1981; Timmermann, 1993), a feature that can substantially amplify the effects of economic cycles. This volatility affects the capital firms can raise through IPOs, shaping employment dynamics through two main channels. First, firms that go public during downturns may secure only limited capital, potentially missing a critical window to establish a strong foundation for sustained employment growth.¹ Second, at the aggregate level, such volatility may induce firms to postpone IPOs until conditions improve, contributing to the procyclicality of IPO activity and, consequently, amplifying aggregate employment fluctuations over the business cycle.

This paper investigates the mechanisms underlying the cyclical nature of IPO activity and its implications for aggregate employment fluctuations. Using Compustat data, I document that the number of IPOs in the United States is highly procyclical, and that firms going public at different phases of the business cycle follow distinct employment growth trajectories. Specifically, firms that go public during contractions exhibit significantly slower post-IPO employment growth than those going public during expansions, with the disparity emerging in the second year and reaching roughly 20 percentage points by year ten. This gap is closely linked to the amount of capital raised at the time of listing. Motivated by these findings, I develop a heterogeneous firm dynamics model with endogenous IPO decisions to examine the mechanisms driving IPO cyclicity. The model incorporates aggregate productivity shocks that affect the stochastic discount factor and interest rates, which in turn influence firms’ IPO timing through their effects on capital injection and the demand for external finance. I calibrate the model to the U.S. economy over the 2000–2019 period and use it to

¹Sedláček and Sterk (2017) provide similar evidence in the context of firm entry, showing that firms entering during recessions experience persistently lower employment growth due to constrained initial expansion.

quantify the macroeconomic impact of IPO cyclicalities on aggregate employment.

Going public is an endogenous decision, made alongside other firm choices such as investment and exit. The IPO process entails substantial fixed costs, including underwriting fees and legal and accounting expenses. In exchange for these costs, firms issue shares to public investors during the IPO, receiving capital that becomes part of the firm’s equity base. This capital can be used to finance new projects, reduce debt, or expand operations. One structural change accompanying an IPO is the requirement to distribute a portion of future earnings to the public shareholders in the form of dividends. Another key distinction of public firms lies in mandatory information disclosure: investor protection regulations require publicly listed firms to maintain greater transparency, which increases ongoing operating costs (Breuer and Breuer, 2023) but also facilitates access to external debt (Schenone, 2010). Weighing these costs and benefits, firms endogenously determine the timing of their IPOs. In this paper, I treat the transition from private to public status as a one-time structural shift.

I find that the procyclicality of IPO activity amplifies the effects of aggregate productivity shocks on employment dynamics through two main channels. First, a lower share of public firms exacerbates capital misallocation, increases production inefficiency, and reduces average firm size. Public firms benefit from greater transparency due to mandatory disclosure requirements, which relaxes their borrowing constraints and enables faster growth. As a result, a lower share of public firms worsens capital allocation across the economy. Second, a lower propensity to go public reduces the expected future value of potential entrants. Since new firms typically enter with limited initial capital, constrained access to public equity—or continued exposure to tighter borrowing constraints in the absence of going public—deters entry. Limited access to external finance after entry reduces the incentive to enter at all, weakening firm dynamism and long-run employment growth.

Related Literature. My paper connects to three branches of the literature. First, my paper relates to the literature on examining heterogeneous firm dynamics models over the business cycle. Many studies focus on the cyclical entry and exit margins, building upon the model developed by Hopenhayn (1992). For example, Lee and Mukoyama (2015) highlights the significant role of cyclical entry costs in explaining the cyclical entry rates observed in the data. Clementi and Palazzo (2016) extend the model

to include capital, finding that the cyclical entry margin is an important mechanism for the propagation of aggregate shocks. While the aforementioned papers are about the extensive margin, Moreira (2016) and Sedláček and Sterk (2017) shed light on the intensive margin of entrants, showing that entrants during adverse economic conditions are not only fewer but also characterized by lower growth potential. This paper extends Clementi and Palazzo (2016) by incorporating an endogenous IPO decision to explore its role in the propagation of aggregate shocks. By examining IPO decisions influenced by business cycles, this study addresses the transition of firms from private to public status and its impact on firm dynamics, a dimension not explored in previous research.

Second, my paper connects to the literature on firms' financial choices in the presence of financial frictions. The impact of financial friction on firm dynamics is cyclical through the channel of firms' investment (Kiyotaki and Moore, 1997; Khan and Thomas, 2013), but most analyses are limited to debt financing. In practice, firms use both debt and equity financing, with the latter being procyclical, unlike debt finance (Jermann and Quadrini, 2012). Considering this, Di-Nola (2016) shows that models including equity financing may reveal that the impact of financial frictions on firm dynamics has been overestimated. The firm's external financing choice is thus crucial for understanding firm dynamics over the business cycle, which includes the endogenous IPO decision, choosing equity issuance over debt financing. A distinctive feature in this paper is the assumption that equity issuance is a one-time decision corresponding to the transition from private to public status, unlike the recurring debt-equity choice in other models.

Third, my paper relates to the literature studies the firms' IPO decisions. While many studies provide empirical insights into the firm's cyclical decision on IPOs (Jovanovic and Rousseau, 2004; Helwege and Liang, 2004; Tran and Jeon, 2011; Chemanur and He, 2012; Angelini and Foglia, 2018), theoretical insights are comparatively scarce. Some papers offer theoretical models integrating the IPO decision to explore the long-term trend of IPOs rather than business cycles. González (2020) quantitatively analyze the differential impact of tax environments on private and public firms, explaining the surge in IPOs during the 1990s. This model shares similarities with Midrigan and Xu (2014), which examines capital misallocation through a one-time equity issuance during sector transitions from traditional to modern. Although they do not specifically refer to these sectors as private and public firms, the concept of irreversible transition through the one-time equity issuance is akin to the endogenous

IPO choice. Casella et al. (2022) and Davydiuk et al. (2020) attribute the decline in IPOs to increased costs, such as operating expenses and mandatory disclosure. While this paper focuses on short-term fluctuations over the business cycle, it builds upon these quantitative firm dynamics models.

The remainder of the paper is organized as follows. Section 2 introduces data and the empirical evidence regarding the cyclical nature of going public. Section 3 presents the model, and section 4 documents the results of the quantitative analysis of the model.

2 Empirical Evidence

In this section, I present empirical evidence on IPO cyclicalities, examining IPO frequency over the business cycle, firm composition, and post-IPO performance, including exit patterns and employment growth.

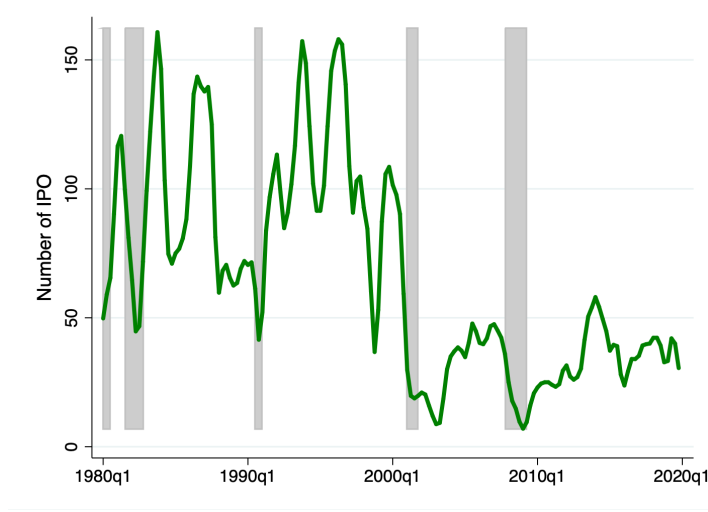
2.1 Data

I use the Compustat/CRSP merged dataset, which combines stock market data from CRSP with firm fundamentals from Compustat for publicly traded U.S. firms. The dataset includes IPO-related information such as the first effective date, IPO date, and capital raised, measured by total stock sales around the IPO year. Following standard practice, I exclude firms in utilities (SIC 4900–4949), finance and insurance (SIC 6000–6999), and public administration (SIC 9000–9999), as well as firms headquartered outside the United States and subsidiaries. I also drop firms that underwent leveraged buyouts or were delisted within three years of listing due to mergers or acquisitions. These restrictions yield a panel of 12,858 firms with 113,745 firm-year observations from 1980 to 2019. All financial variables are expressed in millions of 2012 U.S. dollars, deflated using the Consumer Price Index from the U.S. Bureau of Economic Analysis. Firm-level variables are winsorized at the 1st and 99th percentiles to limit the influence of outliers.

2.2 Cyclicalities of going public

Number of IPOs. The procyclicality of the number of IPOs is well documented (Alti, 2005; Pástor and Veronesi, 2005; Chemmanur, 2010). Figure 1 plots quarterly

Figure 1: Number of firms going public over the business cycle



Notes. The figure presents the number of IPOs in the U.S. based on quarterly IPO dates from Compustat/CRSP. Economic recessions, as defined by the National Bureau of Economic Research, are shaded in gray.

IPO counts, with recession periods shaded in gray. The figure confirms a sharp decline in IPO activity during downturns within the selected sample.² The drivers of IPO cyclicity have been extensively studied. At the micro level, firms are more likely to go public during expansions, when their productivity tends to peak over the cycle (Chemmanur et al., 2010), allowing them to obtain higher valuations at issuance (Benninga et al., 2005). IPO activity also tends to cluster across industries and regions. Alti (2005) attributes this pattern to information spillovers, which reduce IPO-related uncertainty and issuance costs, amplifying the aggregate response to business cycles. At the macro level, IPO frequency is closely linked to broader financial conditions, including interest rates and stock market performance (Tran and Jeon, 2011; Angelini and Foglia, 2018). According to Jovanovic and Rousseau (2004), low interest rates ease access to debt financing, thereby reducing firms' reliance on equity markets and potentially delaying IPOs.

Overall, this evidence suggests that IPO decisions are endogenous responses to both firm-specific characteristics and aggregate conditions. Given that IPOs are shown to boost firm-level employment growth (Borisov et al., 2021), this cyclical pattern implies that the number of firms expanding employment via IPOs fluctuates with the business cycle, potentially amplifying aggregate employment dynamics. Despite its

²Other declines coincide with stock market performance. See Table 8 in Appendix A.1.

Table 1: Median statistics of firms at IPO

	Expansion	Contraction
Employment (1K)	0.20	0.13
Asset (1M, \$)	2,575	470
Debt-to-asset ratio	0.41	0.44
Sale of stock (1M, \$)	622.6	54.4
relative to Asset	0.35	0.12

Notes. The ‘Contraction’ (‘Expansion’) cohort includes firms that went public between a business cycle peak and trough (trough and peak). The dataset is pooled cross-sectional data of IPO firms from Compustat/CRSP. For more detailed summary statistics, see Table 9 in the appendix.

macroeconomic relevance, this amplification channel has received limited attention in the literature.

IPO cohort. I examine whether firms that go public at different phases of the business cycle differ systematically in their characteristics and post-IPO growth trajectories. To do so, I classify firms into two cohorts based on IPO timing: those that go public during economic contractions (from peak to trough) and those during expansions (from trough to peak).³ Descriptive statistics in Table 1 show that, on average, contraction IPO firms are smaller in both employment and total assets. The two cohorts exhibit comparable capital structure, with average debt-to-asset ratios around 0.4. However, contraction IPO firms raise significantly less capital from public equity markets than their expansion-period counterparts: the latter raise approximately 35 percent of their assets at IPO, compared to just 12 percent for the contraction cohort. This gap persists after controlling for firm-specific characteristics (see Appendix A.3 for details).

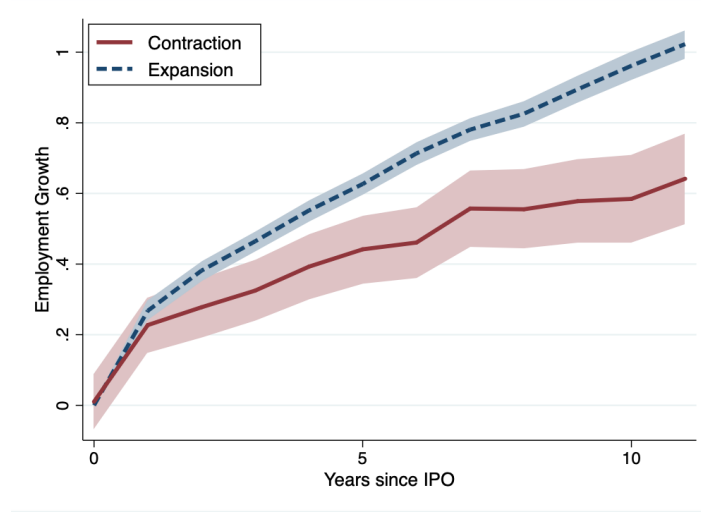
To analyze post-IPO employment growth by age for each cohort, I run a non-parametric regression

$$\Delta^{ipo} N_{it} = \beta_{a,c} Age_{it} \times Cohort_i^{ipo} + \Gamma \mathbf{X}_i \mathbf{0} + \gamma Y_{t-1} + \epsilon_{it}. \quad (1)$$

Here, $\Delta^{ipo} N_{it}$ denotes the cumulative employment growth rate from the year of IPO, calculated as $\ln(N_{it}) - \ln(N_{i0})$ where $t = 0$ represents the IPO year. The coefficient $\beta_{a,c}$

³Business cycle dates are defined by the National Bureau of Economic Research.

Figure 2: IPO cohort effect on post-IPO employment growth

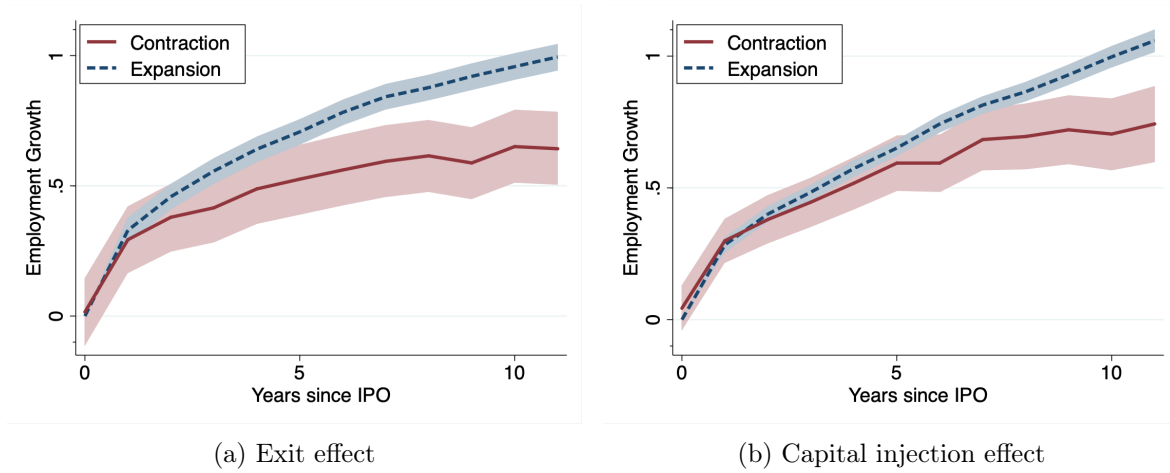


Notes. The figure displays cumulative post-IPO employment growth for cohorts classified by IPO timing within the business cycle in an unbalanced panel. The ‘Contraction’ (‘Expansion’) cohort includes firms that went public between a business cycle peak and trough (trough and peak). These results are based on the non-parametric regression specified in Equation (1), controlling for initial employment, sales per worker, pre-IPO capital intensity, and debt-to-asset ratio at the IPO year, as well as industry and regional fixed effects; Source: Compustat/CRSP.

captures the cohort effect, representing the average employment growth by IPO age (Age_{it}), defined as a dummy for the number of years since the IPO, and differentiated by the cohorts formed during economic expansions and contractions ($Cohort_i^{ipo}$). Control variables $\mathbf{X}_i\mathbf{0}$ include firm i ’s initial employment, sales per worker, pre-IPO capital intensity (gross value of property, plant and equipment), and liabilities (debt-to-asset ratio) at the year of IPO, as well as industry and regional fixed effects. Additionally, Y_t represents the detrended log of real GDP, accounting for varying economic conditions encountered by expansion and contraction cohorts post-IPO.

Figure 2 plots post-IPO employment trajectories, emphasizing the cohort effect. Firms that go public during contractions grow more slowly than those that do so during expansions. By year six, the average employment gap exceeds 10 percentage points, widening to 20 percentage points by year ten. This pattern highlights the importance of IPO timing for subsequent firm growth. To explore the sources of this growth gap, I first restrict the sample to a balanced panel of firms that survive at least ten years post-IPO, accounting for differential exit rates across cohorts (see Table 10

Figure 3: IPO cohort effect on post-IPO employment growth: extended analysis



Notes. The figures extend the analysis from Figure 2 displaying cumulative post-IPO employment growth. Panel (a) based on a balanced panel, examines the impact of heterogeneous exit patterns between contraction and expansion cohorts. Panel (b) controls for the amount of capital raised at the IPO, measured by the sale of stock in the IPO year, to assess the influence of varying levels of capital injection at IPO.

in the appendix). Panel (a) of Figure 3 shows that, although controlling for exit narrows the gap somewhat, a difference of at least 10 percentage points persists at year ten. Second, since post-IPO growth is partly financed through internal resources (Tran and Jeon, 2011), I examine whether variation in capital raised at the time of IPO contributes to the observed disparity. Specifically, I extend the non-parametric regression in Equation (1) by including the amount of capital raised at IPO as an additional control. Panel (b) of Figure 3 shows that this adjustment substantially reduces the growth gap: by year eight, the difference is no longer statistically significant, and by year ten, it is nearly halved. These results suggest that procyclical variation in capital raised at IPO is a key driver of the long-run persistent employment growth gap between contraction and expansion cohorts.

One cannot rule out the possibility of omitted variable bias. For example, capital injection may proxy for unobserved firm productivity, which could itself drive post-IPO employment growth. Consequently, the empirical findings in this section may reflect selection bias, as more productive firms might be more likely to go public during expansionary periods. This raises a central question: are productive firms endogenously timing their IPOs to coincide with favorable macroeconomic conditions? Or, alternatively, as previously suggested, does the lower capital injection during downturns

compress the subsequent growth of IPO cohorts? To disentangle these mechanisms and assess their aggregate implications, I develop and estimate a quantitative firm dynamics model with endogenous IPO decisions.

3 Model

Time is discrete, $t = 1, 2, \dots$, and the horizon is infinite. There are two types of firms, privately held and publicly traded. Firms enter the market as private and may transition to public status through an IPO. For tractability, the transition is assumed to be irreversible.⁴ Both types of firms produce a homogeneous good at time t according to the production function

$$y_{it} = z_t s_{it} (k_{it}^\alpha l_{it}^{1-\alpha})^\eta. \quad (2)$$

Output y_{it} depends on two components of productivity: an aggregate component z_t and a firm-specific s_{it} . Given these, firms use two inputs—labor l_{it} and predetermined capital k_{it} . The parameter $\alpha \in (0, 1)$ governs the capital share, while $\eta \in (0, 1)$ determines the degree of decreasing returns to scale.

The aggregate component of productivity z_t evolves according to a first-order autoregressive process

$$\log(z_{t+1}) = \rho_z \log(z_t) + \varepsilon_t, \quad (3)$$

with $\varepsilon_t \sim \mathcal{N}(0, \sigma_z)$. In both public and private firms, firm-specific productivity s_{it} evolves stochastically according to

$$\log(s_{i,t+1}) = \rho_s \log(s_{it}) + \nu_{it}, \quad (4)$$

with $\nu_{it} \sim \mathcal{N}(0, \sigma_s)$. The shocks ε_t and ν_{it} are assumed to be orthogonal, ensuring independent variation in aggregate and idiosyncratic productivity. The initial productivity of new entrants, s_0 is specified below. For notational simplicity, the subscript i

⁴While firms may later delist and return to private status, delistings do not exhibit a clear cyclical pattern. Over the past 40 years, the delisting rate has remained relatively stable, with the exception of a temporary spike following regulatory changes in the early 2000s. Given this, delisting can be modeled as an exogenous shock. Since such shocks are unlikely to affect the core implications of the model, I focus on IPO timing and treat it as an irreversible decision.

will be omitted for firm-specific variables in the remainder of the exposition.

Aggregate Fluctuation. Aggregate uncertainty affects firms' behavior. Following Clementi and Palazzo (2019), I assume that firms assess future cash flows using a stochastic discount factor, defined as

$$M(z_t, z_{t+1}) \equiv \beta \exp(\phi_0 \log(z_t) + \phi_1 \log(z_{t+1})) , \quad (5)$$

where the subjective discount factor, $\beta > 0$ is the time discount factor, and the parameters ϕ_0 and ϕ_1 represent the sensitivity to aggregate states. The risk-free rate is defined as the inverse of the expected value of the stochastic discount factor which reduces to⁵

$$R_t = \frac{1}{\beta} \exp \left(-z_t(\phi_0 + \rho_z \phi_1) - \frac{\phi_1^2 \sigma_z^2}{2} \right) . \quad (6)$$

With the parameter restriction $\phi_1 \in \left(-\frac{\phi_0}{\rho_z}, 0 \right)$, the risk-free rate is countercyclical.⁶

IPO Decision and Financial Friction. Private firms decide whether to go public, incurring a one-time fixed cost κ associated with the IPO process. Through an IPO, firms offer a χ share of their future dividends to public investors, with the price of each share, denoted $p(k_t, b_t, s_t, z_t)$, determined by demand from a large pool of investors who apply the same stochastic discount factor $M(z_t, z_{t+1})$ as the firms. The price depends on the aggregate state z_t and firm-specific factors, including productivity s_t and capital structure k_t and b_t . Specifically, by selling a fraction χ of its future earnings, the firm obtains funding equal to $\chi p(k_t, b_t, s_t, z_t)$, where $p(\cdot)$ denotes the firm's valuation at IPO. Following the IPO, public firms distribute a fraction χ of their dividends to outside investors in each period. Thus, firms face an intertemporal trade-off between the immediate benefit of a capital injection and the future cost of sharing earnings with shareholders.

Firms operate under financial constraints, issuing risk-free debt b_t subject to a

⁵Derivation is provided in the appendix B.1.

⁶See Clementi and Palazzo (2019) and Beaudry and Guay (1996), among others, for detailed discussions on the countercyclicality of the risk-free rate.

collateral constraint

$$b_t \leq \theta k_t, \quad (7)$$

where $\theta = \{\theta^{pb}, \theta^{pr}\}$ denotes the borrowing limit as a share of capital, and may differ between public and private firms. This difference captures variation in transparency and access to credit markets. For public firms, financial frictions are mitigated by mandatory disclosures (e.g., SEC filings), which enhance transparency and facilitate access to external debt (Schenone, 2010). As a result, public firms likely face a more relaxed borrowing constraint, i.e., $\theta^{pb} \geq \theta^{pr}$. Greater transparency, however, comes at a cost. Public firms incur higher operating expenses, such as auditing and reporting fees, denoted by $f = \{f^{pb}, f^{pr}\}$, with $f^{pb} \geq f^{pr}$. To remain agnostic in the theoretical setup, I do not impose parametric restrictions on the relative magnitude of borrowing constraints or operating costs across firm types. Instead, I allow the calibration to determine whether the expected relationships hold empirically.

Incumbents' Problem. Following the transition, public firms distribute dividends to shareholders each period, corresponding to the share sold at the IPO. Consequently, the public firm retains a $1 - \chi$ fraction of the total dividends in the post-IPO period. Additionally, operating as a public firm entails higher costs. Therefore, in each period t , the public firm aims to maximize the $1 - \chi$ share of the discounted value of its dividend stream⁷ subject to the following budget constraint

$$d_t + k_{t+1} - b_{t+1} = y_t - w l_t - (1 + r(z_t))b_t + (1 - \delta)k_t - f^{pb}. \quad (8)$$

Here, d_t denotes dividends, which are subject to non-negativity constraints ($d_t \geq 0$). The variables w , $r(z_t)$, and f^{pb} represent wage, interest rate, and operating cost, respectively.

Private firms maximize the entire share of the discounted value of its dividend stream, subject to budget constraint

$$d_t + k_{t+1} - b_{t+1} = y_t - w l_t - (1 + r(z_t))b_t + (1 - \delta)k_t - f^{pr} + \{\chi p(k_t, b_t, s_t, z_t) - \kappa\} \xi_{ipo}, \quad (9)$$

⁷For the public firm's decision, it is irrelevant whether they maximize the dividend stream or its discounted value by $1 - \chi$. However, this distinction is crucial for private firms, which compare the value function of remaining private with that of becoming a public firm.

where $\xi_{ipo} \in \{0, 1\}$ is a binary indicator of the IPO decision. This equation elucidates that firms going public incur a fixed cost κ and receive capital injection $\chi p(k_t, b_t, s_t, z_t)$ by issuing equity. As above, f^{pr} represents the operating cost.

Exit and Entry. Both private and public firms decide whether to exit after their production and prior to making investment. Upon exit, firms liquidate the internal cash flow and depreciated capital, net of outstanding debt, which reduces to

$$d_t^{x,pr/pb} = y_t - wl_t - (1 + r(z_t))b_t + (1 - \delta)k_t - f^{pr/pb}. \quad (10)$$

For public firms, a χ share of dividends is allocated to investors, and accordingly, an equivalent share of liquidation proceeds is distributed to them upon exit. Consequently, exiting public firms retain only $(1 - \chi) d_t^{x,pb}$.

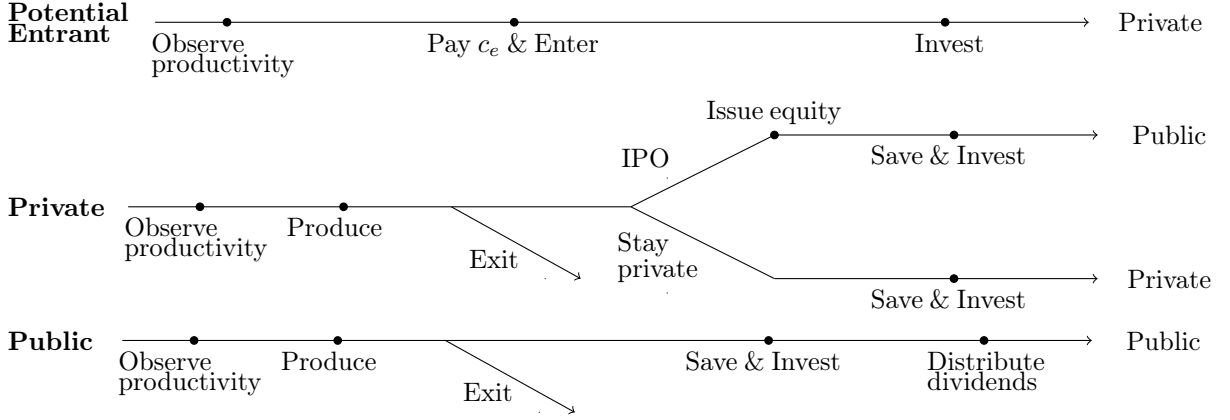
Each period, a large number of potential entrants observe the current aggregate productivity level z_t and draw their initial idiosyncratic productivity s_0 from a log-normal distribution. Entrants are endowed with initial capital k_0 , and debt is unavailable at the entry stage. They can freely enter as a private firm but requires paying a fixed cost $c_e > 0$. Entrants compare this cost to the expected value of operating in the next period. If the expected continuation value exceeds the entry cost, they choose to enter. Upon entry, they invest and begin production in the following period as private firms.

Price of share. Selling a fraction χ of equity at IPO entitles investors to an equivalent share of dividends in every subsequent period. This determines the share price $p(k_t, b_t, s_t, z_t)$ according to

$$\chi p(k_t, b_t, s_t, z_t) = \mathbb{E}_t \left[M(z_t, z_{t+1}) \max \left\{ \chi d_{t+1} + \chi p(k_{t+1}, b_{t+1}, s_{t+1}, z_{t+1}), \chi d_{t+1}^{x,pb} \right\} \right]. \quad (11)$$

Since dividends are paid starting in the period following the IPO, they are discounted using the stochastic discount factor $M(z_t, z_{t+1})$. Here, $d_{t+1} = d(k_{t+1}, b_{t+1}, s_{t+1}, z_{t+1})$ denotes dividends in the next period, derived from the firm's post-IPO budget constraint (8). The term $d_{t+1}^{x,pb}$ denotes the liquidation value upon exit, as defined in (10). This formulation reflects the investor's expected payoff from holding a χ equity share in a public firm, accounting for both ongoing dividends and the possibility of firm exit. The pricing equation introduces a fixed-point problem, which will be further specified

Figure 4: Timing of firms' problem



below using the public firm's value function.

Recursive formulation. The time subscript is suppressed in the following recursive formulation. Firms discount future value by $M(z, z')$.

Figure 4 illustrates the timeline of the firms' problem. After observing the aggregate state z and drawing the initial productivity s_0 , entrants pay the entry cost c_e , make an initial investment decision, and subsequently become a private firm. Similarly, incumbent firms begin with observing an aggregate shock z and idiosyncratic productivity s . Subsequently, firms hire labor maximizing profit given the state of productivity s and z and pre-determined capital k as

$$\begin{aligned}
 l(k, s, z) &= \arg \max \quad sz(k^\alpha l^{1-\alpha})^\eta - wl \\
 &= \left(\frac{(1-\alpha)\eta}{w} sz k^{\alpha\eta} \right)^{\frac{1}{1-(1-\alpha)\eta}}.
 \end{aligned} \tag{12}$$

Each period begins with the payment of relevant costs, including wages, interest, and operating expenses. Incumbent firms then decide whether to exit the market. If a firm exits, its assets are liquidated. Shareholders of public firms—those who invested at the IPO—receive a fraction χ of the liquidation value, while the remaining portion (and the entire liquidation value of private firms) is recycled into entry. Next, private firms choose whether to go public by issuing equity through an IPO. Firms undertaking an IPO sell shares and internalize the resulting capital injection. Subsequently, both private and public firms make investment and financing decisions, determining how to allocate internal and external funds toward investment. Finally, public firms distribute

dividends from residual profits.⁸

The period begins with the payment of relevant costs such as wages, interest, and operating expenses. Incumbents then decide whether to exit the market, followed by the decision of private firms to go public through an IPO. Firms undertaking an IPO issue equity, which involves selling shares and internalizing the capital injection. Subsequently, both private and public firms make investment and financing decisions, determining how to utilize internal and external capital for these investments. Finally, public firms distribute dividends from the residual profits.

The investment and financing decisions proceed as follows: in each period, firms choose capital investment k' and determine how to finance it by allocating between internal cash flows and new debt b' . Internal cash flows are predetermined by the realized productivity components s and z , observed at the beginning of the period, as well as by the firm's inherited capital k and debt b . Consequently, when desired investment k' exceeds internal cash flows, the firm must finance the shortfall through debt issuance, implying that the choice of k' determines b' . To simplify the dynamic problem, I define net worth as the difference between capital and debt, $a \equiv k - b$. The firm's intertemporal decision can then be recast as a choice over future net worth a' , with an associated intra-period problem that solves for the optimal pair (k', b') given a' :

$$\begin{aligned} \pi' &= \max_{k', b'} \int \int s' z' (k'^{\alpha} l'^{1-\alpha})^{\eta} - w l' - (1 + r(z')) b' + (1 - \delta) k' dH(s'|s) dG(z'|z), \\ \text{s.t. } k' - b' &= a', \\ b' &\leq \theta k', \\ l'(k', s', z') &= \left(\frac{(1 - \alpha)\eta}{w} s' z' k'^{\alpha\eta} \right)^{\frac{1}{1-(1-\alpha)\eta}}. \end{aligned}$$

which simplifies to an optimal capital choice k'^* as

$$k'^* = \min \left(\frac{a'}{1 - \theta}, \Phi \left(\frac{\int \int \left(\frac{s' z'}{w^{(1-\alpha)\eta}} \right)^{\frac{1}{1-(1-\alpha)\eta}} dH(s'|s) dG(z'|z)}{\int r(z') dG(z'|z) + \delta} \right)^{\frac{1-(1-\alpha)\eta}{1-\eta}} \right),$$

⁸Distributed dividends for public firms are determined as residual profits after investment and financing decisions. This assumption captures the observed high reinvestment rates in the years following an IPO.

where $\Phi = (\alpha\eta)^{\frac{1-(1-\alpha)\eta}{1-\eta}}((1-\alpha)\eta)^{\frac{(1-\alpha)\eta}{1-\eta}}$, and debt as $b'^* = k'^* - a'$. As a result, profit function can be expressed as a function of current net worth and productivity (a, s, z) as follows

$$\pi(a, s, z) = \Omega(szk^{*\alpha\eta})^{\frac{1}{1-(1-\alpha)\eta}} - (r(z) + \delta)k^* + (1 + r(z))a, \quad (13)$$

where $\Omega = (1 - (1 - \alpha)\eta) \left(\frac{(1-\alpha)\eta}{w} \right)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}}$. Hence, the forthcoming dynamic optimization problems are represented as state vector $\mathcal{S} = (s, z, a)$ and a control variable a' that maximizes the value functions.

The public firms' problem can thus be recursively formulated as

$$V^{pb}(\mathcal{S}) = \max\{\tilde{V}^{pb}(\mathcal{S}), V^{x,pb}(\mathcal{S})\}, \quad (14)$$

$$\tilde{V}^{pb}(\mathcal{S}) = \max_{a'} (1 - \chi)d + \mathbb{E}[M(z, z')V^{pb}(\mathcal{S}')|\mathcal{S}], \quad (15)$$

$$\text{s.t. } d + a' = y - wl - (r(z) + \delta)k - f^{pb} + (1 + r(z))a,$$

$$k \leq \frac{a}{1 - \theta^{pb}},$$

$$d \geq 0,$$

$$V^{x,pb}(\mathcal{S}) = (1 - \chi)[\pi(\mathcal{S}) + (1 + r(z))a]. \quad (16)$$

Public firms decide whether to continue operating or exit based on the value function in (14), where the continuation and exit values are defined in (15) and (16), respectively. The value of operating reflects the $(1 - \chi)$ share of the stream of dividends accruing to the original firm owners, as described above. This value is subject to the budget constraint in (8), the optimal investment choice from the intra-period problem in ??, and the non-negativity constraint on dividends. If the firm exits, it liquidates its current cash flow, which consists of current-period operating profit ($\pi(\mathcal{S}) = y - wl - (r(z) + \delta)k - f^{pb}$) and unused savings from the previous period. A fraction χ of the liquidation value is distributed to outside shareholders. This setup ensures that, from the investors' perspective, no resources are destroyed upon the firm's exit they hold shares.

Private firms solve the following problem:

$$V^{pr}(\mathcal{S}) = \max\{V^{x,pr}, \tilde{V}^{pr}, \tilde{V}^{ipo}\}, \quad (17)$$

$$\tilde{V}^{pr}(\mathcal{S}) = \max_{a'} d + \mathbb{E}[M(z, z')V^{pr}(\mathcal{S}')|\mathcal{S}], \quad (18)$$

$$\text{s.t. } d + a' = y - wl - (r(z) + \delta)k - f^{pr} + (1 + r(z))a,$$

$$k \leq \frac{a}{1 - \theta^{pr}},$$

$$d \geq 0,$$

$$\tilde{V}^{ipo}(\mathcal{S}) = \max_{a'} d + \mathbb{E}[M(z, z')V^{pb}(\mathcal{S}')|\mathcal{S}], \quad (19)$$

$$\text{s.t. } d + a' = y - wl - (r(z) + \delta)k - f^{pr} + (1 + r(z))a + \chi p(\mathcal{S}) - \kappa,$$

$$k \leq \frac{a}{1 - \theta^{pb}},$$

$$d \geq 0,$$

$$\chi p(\mathcal{S}) = \frac{\chi}{1 - \chi} \mathbb{E}[M(z, z')V^{pb}(\mathcal{S}')|\mathcal{S}], \quad (20)$$

$$V^{x,pr}(\mathcal{S}) = \pi(\mathcal{S}) + (1 + r(z))a. \quad (21)$$

Private firms make both exit and IPO decisions as specified in equation (17). The exit value for private firms, given in equation (21), coincides with that of public firms when $\chi = 0$, except that the operating profit is adjusted for private firms by deducting their specific operating costs, i.e., $\pi(\mathcal{S}) = y - wl - (r(z) + \delta)k - f^{pr}$. A private firm chooses to go public if the value of doing so, as defined in (19), exceeds the value of remaining private, given in (18). The value function for going public incorporates the future value of operating as a public firm, reflecting the transition from private to public status that occurs immediately. Under the assumption that financing decisions follow the IPO decision, firms undertaking an IPO solve the intra-period problem subject to the public-sector collateral constraint under θ^{pb} . Equation (19) embeds the price function defined in (11), which is reformulated using the recursive value function of public firms.⁹

Finally, potential entrants choose to enter if the expected value of entry,

$$V^e(\mathcal{S}_0) = \max_{a'} -a' + \mathbb{E}[M(z, z')V^{pr}(\mathcal{S}')|\mathcal{S}_0], \quad (22)$$

⁹This formulation implies that, at the time of the IPO, the private owner retains the full value of the firm. From the following period onward, however, the owner's stake is reduced to $1 - \chi$, with the remaining χ allocated to outside investors. The derivation is provided in Appendix B.4.

exceeds the entry cost c_e .

Recursive Competitive Equilibrium For given initial distribution of Γ_0^{pr} and Γ_0^{pb} , a recursive competitive equilibrium consists of (i) value functions V^{pr} , V^{stay} , V^{ipo} , V^{pb} , $\tilde{V}^{pb}$, and $V^{x,j}$; (ii) policy functions $a^{j'}(a, s, z)$, $k^{j'}(a, s, z)$, $b^{j'}(a, s, z)$, $l^j(a, s, z)$ where $j \in \{pr, pb\}$, and IPO decision $\xi(a, s, z)$; and (iii) the measure of exiters $\{\Gamma_t^{x,j}\}_{t=1}^\infty$, incumbents $\{\Gamma_t^j\}_{t=1}^\infty$ where $j \in \{pr, pb\}$, and entrants $\{\Gamma_t^e\}_{t=1}^\infty$ such that, for all $t \geq 0$:

- (i) V^{pb} , $\tilde{V}^{pb}$, $V^{x,pb}$, $a^{pb'}(a, s, z)$, $k^{pb'}(a, s, z)$, $b^{pb'}(a, s, z)$, $l^{pb}(a, s, z)$ solve the public firm's problem described in (14)-(16).
- (ii) V^{pr} , V^{stay} , V^{ipo} , $V^{x,pr}$, $a^{pr'}(a, s, z)$, $k^{pr'}(a, s, z)$, $b^{pr'}(a, s, z)$, $l^{pr}(a, s, z)$, $\xi(a, s, z)$ solve the private firm's problem described in (17) - (21).
- (iii) $V^e \geq c_e$ described in (22) solves the entrants' problem.
- (iv) Measures evolve according to the law of motion:

$$\begin{aligned}\Gamma^{pb'} &= \Gamma^{pb} - \Gamma^{pb,x} + \xi \Gamma^{pr}, \\ \Gamma^{pr'} &= (1 - \xi)(\Gamma^{pr} - \Gamma^{pr,x}) + \Gamma^e.\end{aligned}$$

4 Quantitative Analysis

4.1 Calibration

The model is calibrated based on U.S. non-financial firm data from the first quarter of 2000 to the fourth quarter of 2019.¹⁰ One period is equivalent to a quarter, and each production unit corresponds to a firm in the dataset.

Table 2 presents the assigned parameter values. The returns to scale parameter η is set to 0.88. This value is consistent with the range between 0.83 and 0.91 that was proposed by Lee (2005). The value of 0.88 ensures that the returns to scale are decreasing, which is a standard assumption in the firm dynamics literature. The capital share α is set at 0.3, which results in a labor share of 0.609. The depreciation rate δ is assigned a value of 0.025, which is equivalent to one-quarter of the commonly utilized

¹⁰The data from the period after 2000 is chosen due to the structural break in IPO dynamics observed in the late 1990s.

Table 2: Assigned parameters

Parameter	Meaning	Value
η	Returns to scale	0.88
α	Capital share	0.30
δ	Depreciation rate	0.03
ρ_z	Persistence aggregate shock	0.95
σ_z	SD aggregate shock	0.007
χ	Equity share sold at IPO	0.10

annual rate of 0.1, as observed in the existing literature. Aggregate productivity shocks are assumed to evolve according to $\rho_z = 0.95$ and $\sigma_z = 0.007$, following Clementi and Palazzo (2019). The equity share sold at IPO, denoted by χ , is estimated using the dividend payout ratio from Compustat, defined as dividends per share divided by earnings per share.¹¹ In the data, this ratio ranges from 0.05 to 0.15 and tends to increase with firms' net worth. However, since its variation does not exhibit a systematic relationship with aggregate economic conditions, I calibrate χ to the sample mean of 0.1.

Table 3: Externally calibrated parameters

	Meaning	Value	Target	Data	Model
β	Time discount	0.97	Mean interest rate	0.004	0.004
ϕ_0	Stochastic discount factor	28.59	SD interest rate	0.009	0.008
ϕ_1	Stochastic discount factor	-30.90	Sharpe ratio	0.22	0.22

Table 3 presents the externally calibrated parameter values. The three parameters governing the stochastic discount factor are jointly determined in order to match the first and second moments of the risk-free rate and the mean Sharpe ratio of public firms, following Clementi and Palazzo (2019). In particular, the estimated mean Sharpe ratio is used to match its upper bound, which is expressed as a function of the stochastic discount factor:

$$\overline{SR} = \frac{\sigma(M_{t+1})}{E_t[M_{t+1}]} = \sqrt{\exp(\phi_1^2 \sigma_z^2) - 1}.$$

¹¹Since young firms in Compustat typically do not distribute dividends and instead reinvest earnings, including them may lead to underestimation of the payout ratio. Therefore, I restrict the sample to older firms—those listed for at least 10 years—when calculating this measure.

Thus, ϕ_1 is set to match the target Sharpe ratio, while β and ϕ_0 are calibrated to align with the remaining moments of the risk-free rate. Due to the nonlinearities inherent in the mapping between parameters and moments, two distinct sets of values can be identified that satisfy the specified targets. One set produces a countercyclical risk-free rate, while the other produces a procyclical rate. I adopt the countercyclical specification, matching the target values estimated by Clementi and Palazzo (2019) using Compustat data.

Table 4: Targeted moments

	Meaning	Value	Target	Data	Model
ρ_s	Persistence idio. shock	0.93	AC of log sales	0.63	0.41
σ_s	SD idiosyncratic shock	0.07	SD of log sales	0.35	0.35
θ^{pb}	Borrowing constraint	0.55	Debt-to-asset	0.77	0.77
θ^{pr}	Borrowing constraint	0.35	Debt-to-asset	0.42	0.40
f^{pb}	Operating cost	5.20	Exit rate	0.02	0.03
f^{pr}	Operating cost	1.50	Exit rate	0.09	0.08
κ	IPO fixed cost	170	Emp. share of public	0.33	0.35

Notes. The first two parameters related to idiosyncratic productivity are calibrated using the simulated method of moments. The remaining parameters, which determine the costs and benefits of IPOs, are calibrated to align with observed moments for public and private firms, utilizing data from Compustat and aggregate sources such as Flow of Funds and BDS.

Table 4 reports the calibration of parameters determined internally. The idiosyncratic shock process is calibrated using the Simulated Method of Moments (SMM) to match the first-order autocorrelation and cross-sectional standard deviation of log output among public firms. These moments are estimated from Compustat using quarterly sales data. To construct empirical targets, I restrict the sample to firms with at least 20 quarters of non-missing log sales data. The first-order autocorrelation of log sales is used to capture persistence, while the cross-sectional variance identifies volatility. Year fixed effects are included to control for aggregate shocks, ensuring the estimated moments reflect firm-specific dynamics. The same procedure is applied to simulated data from the stationary distribution of the model for moment comparison.¹²

¹²The model assumes a homogeneous idiosyncratic productivity process across private and public firms, which may contribute to discrepancies in persistence and volatility between the model and the data. Empirical evidence from Davis et al. (2007) suggests that public firms tend to experience more persistent and less volatile productivity shocks than private firms. Therefore, heterogeneous idiosyncratic productivity process between private and public firms would be more appropriate, but I assume they are identical for simplicity in the baseline model, and extend it later.

Table 5: Size distribution of firms

Size	1 to 9	10 to 19	20 to 99	100+
Model	0.61	0.24	0.12	0.03
Private	0.65	0.25	0.10	0.01
Public	0.04	0.13	0.50	0.32
Data	0.77	0.12	0.10	0.02
Public	0.04	0.03	0.17	0.76

Notes. This table presents the firm size distribution by number of employees. The model captures the stationary distribution of firms, while the data shows the average firm share within each category based on BDS, and Compustat for the public firms.

The remaining parameters govern the IPO dynamics and the heterogeneity between private and public firms. The parameter determining the borrowing constraint θ is calibrated by matching the debt-to-asset ratio. This is achieved by utilizing data from Compustat for public firms and aggregate data from the Flow of Funds for private firms.¹³ The discrepancy in collateral constraints between public and private firms directly influences the incentive to pursue an IPO. A greater disparity results in greater benefits from going public, as it determines how easily public firms can access debt financing and realize faster growth. Operating costs are calibrated to match the exit rates of public firms using Compustat data, and private firms using Business Dynamics Statistics (BDS) data. The exit rate for public firms is even more modest than that observed for private firms. This is due to the fact that public firms are larger and more productive, which makes them less likely to exit. Nevertheless, the calibration results confirm that public firms have higher operating costs than private firms, as suggested during the model construction. Finally, the IPO fixed cost κ is calibrated to pin down the employment share of public firms in the economy.

4.2 Model performance

In this subsection, I present four untargeted moments generated by the model and assess their consistency with the corresponding empirical counterparts.

Table 5 presents the firm size distribution by employment in the model’s stationary

¹³In Compustat, debt is calculated as the sum of long-term debt (`dltt`) and debt in current liabilities (`dlc`), excluding negative values, which are likely to reflect cash holdings. Total assets (`at`) are used as the denominator.

equilibrium. The distribution exhibits a right-skewed pattern that is broadly consistent with empirical observations: over 80 percent of firms employ fewer than 19 workers, while only approximately 2–3 percent have more than 100 employees. This replicates the well-documented empirical regularity that a small share of large firms accounts for a disproportionate fraction of aggregate employment, whereas the vast majority of firms remain small. The table further disaggregates the size distribution across private and public firms in the model and compares the public firm distribution with data from Compustat. The skewness in the overall distribution is primarily driven by private firms, which exhibit the familiar right-skewed pattern. In contrast, the distribution of public firms is skewed to the left in the model, indicating that most public firms are relatively large. This distinction is even more pronounced in the empirical data: while in the model approximately one-third of public firms have more than 100 employees, in the Compustat data the corresponding share exceeds three-quarters.¹⁴

Table 6: Untargeted moments

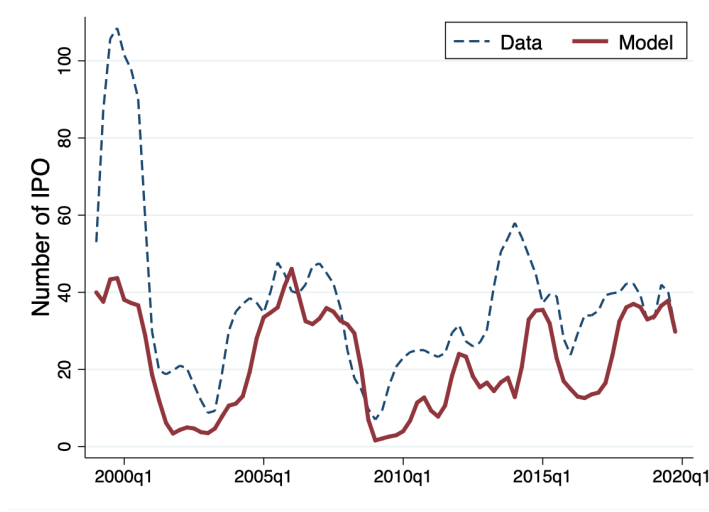
Moment	Model	Data	Source
Growth at IPO	0.82	0.87	Borisov et al. (2021)
IPO effect	0.42	0.37	Borisov et al. (2021)

Notes. In this table, ‘Growth at IPO’ encompasses both the selection effect and the direct effect of the IPO. ‘IPO effect’ isolates the direct impact of going public.

Second, I examine whether the model is capable of replicating the empirical evidence on the contribution of IPOs to employment growth at the firm level. Empirical research by Borisov et al. (2021) documents that IPO firms tend to be fast-growing, and this observed post-IPO employment growth is driven by two distinct channels: (i) a selection effect, whereby firms with already high growth trajectories are more likely to go public, and (ii) an IPO effect, whereby the capital injection from going public enables additional growth. To disentangle these two effects in the model, I simulate a stationary environment and track the employment growth trajectories of firms that choose to go public endogenously. These trajectories are then compared with those of observationally similar firms—matched by size and productivity—that remain private.

¹⁴This discrepancy may be attributed to the insufficient persistence in the productivity process for public firms in the model. A more persistent productivity evolution would allow public firms to sustain higher growth trajectories and better match the concentration of large firms observed in the data.

Figure 5: Simulated number of IPOs



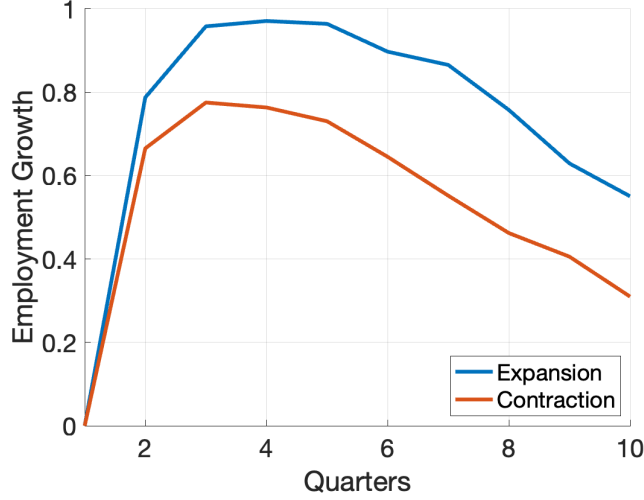
Notes. This figure presents the actual number of IPOs from Compustat/CRSP (blue dashed line) with the simulated IPOs generated by the model (red solid line), using detrended GDP fed in as the input for aggregate productivity in each period.

This counterfactual exercise allows me to isolate the growth attributable purely to the IPO decision from the baseline growth associated with firm characteristics. As shown in Table 6, the model successfully reproduces both the selection and IPO effects documented in the data. That is, firms that undergo an IPO already exhibit higher growth *ex ante* (selection), and they further accelerate employment expansion following the IPO due to enhanced financing capacity (IPO effect). This result supports the model’s micro-level validity in capturing the dual role of IPOs in shaping firm-level dynamics.

Next, Figure 5 compares the number of IPOs observed in the data (blue dashed line) with the corresponding statistics generated by the model (red solid line). In the simulation, the number of IPOs is derived by feeding the detrended GDP series into the model’s aggregate productivity process over the specified period.¹⁵ The simulated IPO dynamics exhibit a distinctly procyclical pattern, closely reflecting key features of the data. In particular, the model successfully captures the sharp decline in IPO activity following major recessionary episodes, such as those occurring after the 2001 dot-com

¹⁵While the theoretical framework interprets aggregate productivity z as total factor productivity (TFP), I find that using the estimated TFP directly fails to reproduce the observed procyclicality and timing of IPO waves. In contrast, detrended GDP more effectively captures broader cyclical forces influencing IPO decisions, including aggregate demand conditions and financial market dynamics. This reduced-form approach enables the model to better replicate the empirical IPO series.

Figure 6: Simulated growth path of expansion/contraction IPO cohort

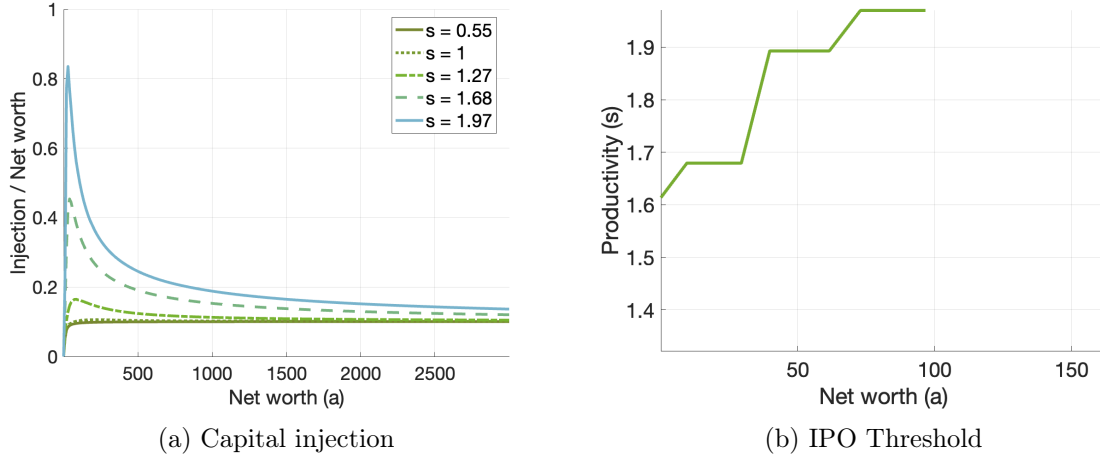


Notes. This figure presents the simulated growth gap between expansion and contraction IPO cohort in the model. Expansion IPO cohort is defined as the IPO firms during the period of $z \geq 1.02$ and contraction cohort is defined as the IPOs during $z \leq 0.98$.

bust and the 2008 global financial crisis. This consistency with the empirical time series provides support for the model's ability to generate realistic extensive-margin responses to aggregate shocks.

Finally, I revisit the heterogeneous post-IPO growth patterns observed in Figure 2, focusing on whether the model can replicate the empirically observed divergence in growth trajectories across IPO cohorts. Specifically, I simulate the model over 1,000 firms and 150 time periods, and use the pooled sample to identify IPO cohorts based on the aggregate productivity level z at the time of going public. I define the expansion cohort as firms that went public during periods of strong aggregate conditions ($z \geq 1.02$), and the contraction cohort as those that went public during downturns ($z \leq 0.98$). The figure illustrates that a persistent growth gap emerges between the two cohorts: firms going public during expansions continue to exhibit stronger post-IPO employment growth relative to their counterparts in contraction periods. While both cohorts converge back to the model's stationary distribution over time, the gap remains substantial in the early post-IPO periods. This pattern qualitatively aligns with the empirical evidence of heterogeneous post-IPO dynamics, suggesting that selection into IPO during different macroeconomic conditions matters for subsequent

Figure 7: Capital injection and IPO decision



Notes. The figure illustrates the amount of capital injection and IPO threshold in a stationary case. Panel (a) displays capital injection normalized by net worth across idiosyncratic productivity (s). Panel (b) shows the IPO threshold in a stationary case ($z = 1$), identifying the region where firms, based on idiosyncratic productivity and net worth, choose to go public when these values exceed the threshold.

performance.¹⁶ In the model, this divergence is driven by the heterogeneous amount of capital injection upon going public, which in turn determines the extent to which borrowing constraints are relaxed. Firms going public during expansions benefit from larger equity inflows, enabling faster growth through accelerated capital accumulation, while those in downturns face relatively more limited financial easing.

4.3 IPO decisions and cyclicalities

Panel (a) of Figure 7 illustrates capital injections at the time of IPO across firms with varying productivity levels, normalized by their net worth. The results reveal a distinct pattern of heterogeneity by firm size and productivity. Specifically, while large firms tend to receive similar levels of capital injection regardless of productivity, small firms show a strong positive relationship between productivity and capital injection. This asymmetry suggests that financially constrained firms—particularly small but highly

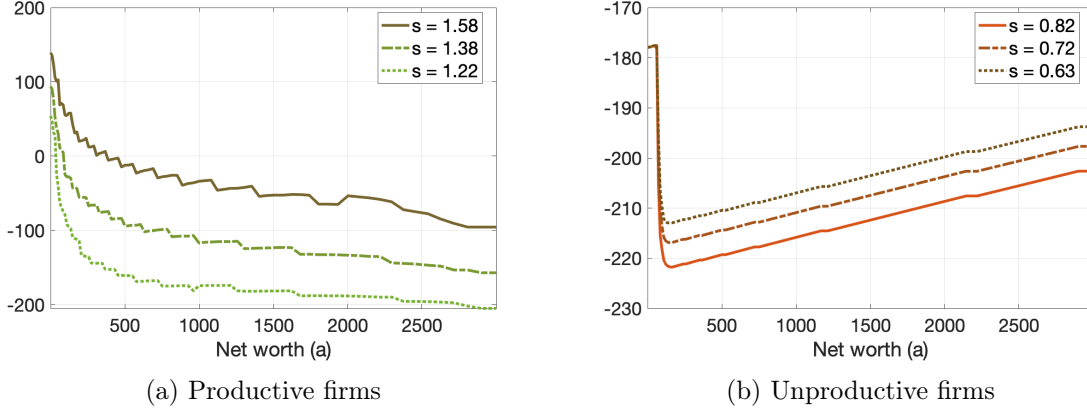
¹⁶However, the relatively fast convergence and limited persistence of the gap in the model appear less pronounced than in the data. One likely explanation is the low persistence of idiosyncratic productivity shocks for public firms in the current calibration. Because productivity does not exhibit strong path dependence, the heterogeneity across IPO cohorts diminishes quickly. Increasing the persistence of firm-level shocks in the public state could help the model better match the empirical duration and magnitude of the growth divergence.

productive ones—are able to raise disproportionately more capital through IPOs. For these firms, the IPO substantially increases expected future value by easing borrowing constraints and improving access to external finance. In anticipation of accelerated growth, investors inject greater amounts of capital at the time of listing, amplifying the post-IPO growth potential of these firms.

Panel (b) of Figure 7 displays the IPO decision thresholds in the model’s stationary equilibrium, plotted over the firm-specific state space: firms’ net worth and idiosyncratic productivity. Firms that lie above the threshold endogenously choose to go public, while those below remain private. The figure reveals that IPO activity is concentrated among firms that are both small and highly productive. This selection pattern reflects two underlying mechanisms: the gains from capital injection and the marginal cost of capital. As shown in Panel (a), capital injections at the time of IPO yield only modest gains—approximately 10 percent of net worth—for firms with low productivity, making the IPO less attractive for this group. Unproductive firms are also less likely to face binding borrowing constraints, reducing their need to access external equity markets in the first place. Conversely, while productive firms have strong growth potential, those with already substantial net worth are typically unconstrained and therefore have less to gain from raising additional external capital. This is reflected in the upward-sloping nature of the IPO threshold: as net worth increases, only firms with correspondingly more productive firms find it optimal to incur the high costs associated with an IPO.

To further examine how firm characteristics shape the decision to go public, Figure 8 presents the value differential between becoming public and remaining private, plotted over firms’ net worth and idiosyncratic productivity. This analysis allows us to assess whether the incentive to go public is monotonically increasing in productivity and decreasing in firm size, or whether more complex patterns emerge. Panel (a) shows the value differential for productive firms, while Panel (b) focuses on unproductive firms. Consistent with earlier findings, only small, productive firms exhibit a positive value differential, indicating that these firms are the primary candidates for going public. In both panels, the curves associated with higher productivity levels lie above those with lower productivity, confirming that the incentive to go public increases with productivity. With respect to firm size, Panel (a) shows that the value differential decreases gradually with net worth, suggesting that the benefits of going public diminish as firms grow larger and become less financially constrained. In contrast, Panel (b) displays

Figure 8: Private firms' value differential

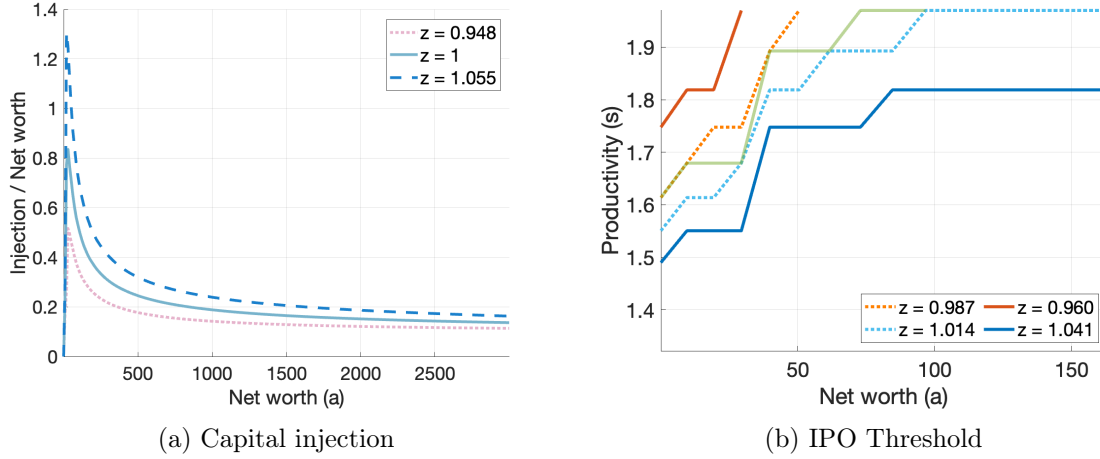


Notes. The figure shows the value differential between firms going public versus staying private, where a positive differential indicates a higher value from going public. The left panel displays the value differential for productive firms ($s = 1.22$ to $s = 1.58$), while the right panel illustrates it for less productive firms ($s = 0.63$ to $s = 0.82$).

a distinct “swoosh” pattern for unproductive firms. Initially, very small unproductive firms benefit from going public due to the relaxation of borrowing constraints. As these firms grow, their constraints are progressively eased, leading to a decline in value differentials. However, once net worth surpasses a certain level (around 73), the value differential begins to rebound. This reversal reflects the fact that larger unproductive firms accumulate sufficient internal resources to absorb IPO-related costs, and the resulting capital injection helps offset the disadvantages of low productivity. Therefore, while the fixed costs of going public discourage most unproductive firms from entering public equity markets, the model suggests that unproductive firms with sufficiently high net worth may still find it beneficial to pursue an IPO. This mechanism may offer an explanation for the empirical presence of some large, low-productivity firms that go public. Nevertheless, under the current calibration, this incentive is not strong enough to translate into an actual IPO decision, and such firms remain private in the model.

Cyclicality of IPOs. The IPO decision exhibits a procyclical pattern. Panel (a) of Figure 9 illustrates how the potential capital injection for a highly productive firm ($s = 1.97$), normalized by net worth, evolves in the nonstationary environment. As shown in the figure, capital injections shift upward with improvements in aggregate productivity, implying that firms are able to raise more external capital during economic expansions. This increase is notably asymmetric: for small firms, capital injections can rise by as

Figure 9: Capital injection and IPO decision in a non-stationary case



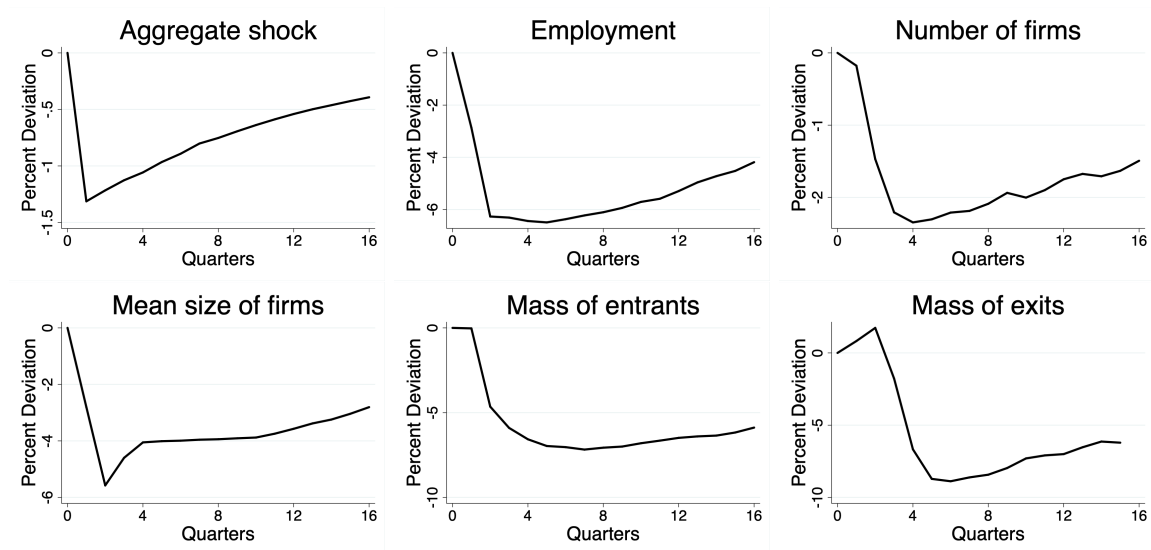
Notes. The figure illustrates the IPO threshold and capital injection in a non-stationary case. Panel (a) shows the normalized potential capital injection at IPO, based on net worth for the most productive firm ($s = 1.97$) across varying aggregate states. In Panel (b), the x-axis is truncated at a net worth of 700, out of a maximum of 3000.

much as 80 percentage points depending on the aggregate state, whereas for larger firms the variation is minimal. This asymmetry is primarily driven by differences in the marginal cost of capital. During periods of high aggregate productivity, more financially constrained firms experience a sharper rise in their marginal cost of capital, thereby benefiting more from the relaxation of borrowing constraints. As investors perceive this expected acceleration in firm growth, the stock price increases, resulting in a larger capital injection at the time of the IPO.

Panel (b) of Figure 9 illustrates how the IPO decision threshold shifts in response to changes in aggregate productivity in the nonstationary setting. As aggregate conditions deteriorate, the IPO threshold moves upward and to the left, indicating that fewer firms choose to go public, and those that do tend to be smaller but more productive. This shift arises because the expected benefits of going public decline in adverse macroeconomic environments. Anticipating a period of weak aggregate conditions, firms expect lower capital injections and reduced marginal gains from relaxing their financial constraints. In such cases, waiting until the conditions improve is more valuable. As a result, only firms with exceptionally high productivity—those that can still justify the costs of going public—endogenously cross the thresholds.¹⁷ This endogenous

¹⁷One mechanism through which firms decide to go public in the nonstationary case relates to the limited persistence of idiosyncratic productivity shocks. Suppose that a firm experiences a large posi-

Figure 10: Response of firm dynamics to a negative productivity shock – Baseline



selection mechanism in the model is consistent with empirical evidence showing that IPO firms tend to be smaller during economic contractions.

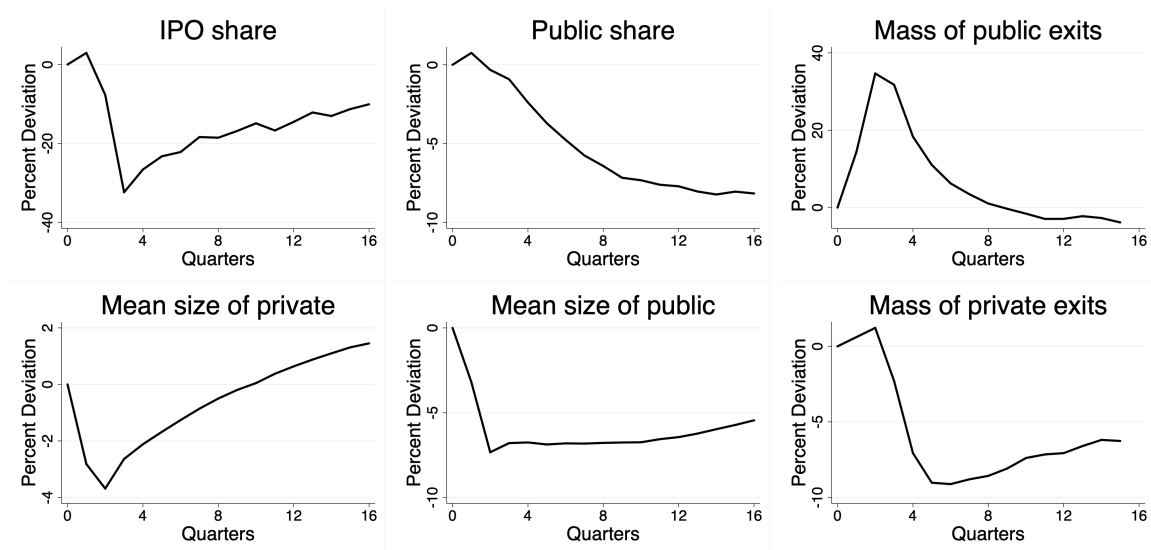
4.4 Impulse responses

I examine impulse response functions following a 1 percent negative MIT shock to aggregate productivity in the baseline model. Starting from the steady state, the shock is introduced at time $t = 1$, and I track the evolution of private and public firm dynamics over the subsequent 16 quarters. This experiment is repeated 1,000 times, and the averages of selected variables are presented.

Figure 10 presents the impulse responses of aggregate employment and firm dynamics—mean size of firms and number of firms. The shock leads to a sharp contraction in employment by approximately 6 percent, followed by a gradual recovery beginning around the sixth quarter. This decline is driven by adjustments along both the extensive and intensive margins. On the extensive margin, the number of firms drops by about 2.5 percent, largely due to a collapse in firm entry immediately after the shock. Over time, the number of incumbents gradually recovers as entry rates rebound

tive shock despite unfavorable macroeconomic conditions. If the firm anticipates that this productivity surge will not persist, delaying the IPO may result in a missed opportunity to accumulate capital, as the associated growth project is time-sensitive and may no longer be viable later. While the expected capital injection could partially recover with macroeconomic improvement, it may nonetheless be lower if the idiosyncratic shock reverses in the meantime.

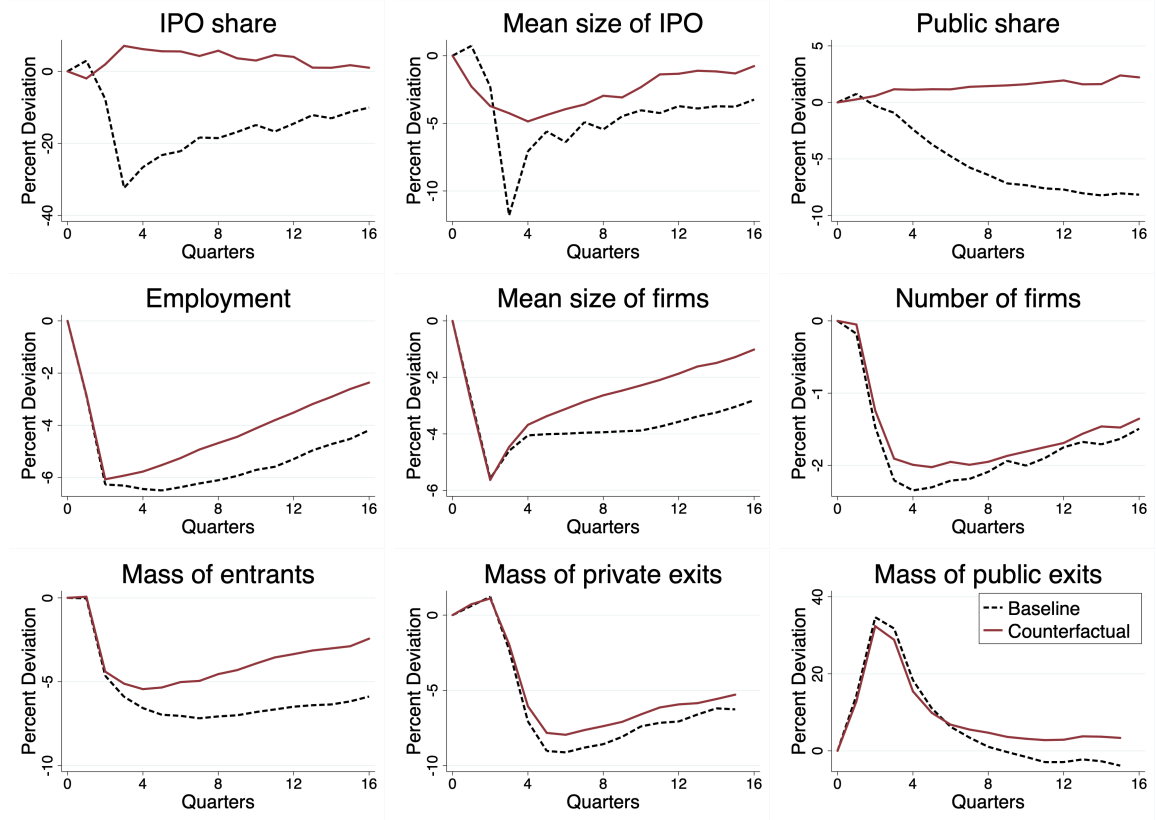
Figure 11: Response of IPO dynamics to a negative productivity shock – Baseline



and exit rates stabilize. On the intensive margin, the average firm size falls by 4 to 5 percent in the immediate aftermath of the shock, further amplifying the decline in aggregate employment.

Figure 11 displays the impulse responses for IPO dynamics and the evolution of private and public firms. The shock induces a sharp contraction in the IPO share—defined as the number of IPOs relative to the stock of private firms—which declines by up to 30 percent. This reduction depresses the entry margin of public firms and lowers their overall share in the firm population to around 8 percent. The decline is further amplified by elevated exit rates among public firms in the first eight quarters after the shock. The dynamics of average firm size diverge notably between private and public firms. The average size of private firms initially contracts by approximately 4 percent but recovers relatively quickly, surpassing its pre-shock level within ten quarters. In contrast, public firms experience a sharper decline of around 7 percent, followed by a protracted recovery phase. Their average size remains below the steady-state level for more than twelve quarters and is still 5 percent lower by the end of the 16-quarter horizon. This asymmetry is driven by the endogenous movement of the IPO threshold, which shifts upward and to the left in response to the shock. As a result, fewer firms transition to public status, and those that do are, on average, smaller. This compositional change delays the recovery in the average size of public firms. Conversely, the average size of private firms recovers more quickly, partly because relatively larger firms remain

Figure 12: Response to a negative productivity shock – Counterfactual I



Notes. The black dashed line represents the response of the baseline model. The red solid line illustrates the response in the counterfactual scenario, in which IPO cyclicalities related to selection and the number of IPOs is shut down.

private and are less likely to exit, which also contributes to a decline in the mass of private firm exits.

To evaluate the role of IPO cyclicalities in the transmission of aggregate shocks to employment, I compare the impulse response functions of the baseline model to those from counterfactual scenarios in which IPO cyclicalities are suppressed. In the first counterfactual, the cyclicalities of both IPO frequency and firm selection are eliminated by exogenously fixing the IPO threshold at its stationary equilibrium value (as shown in Figure 7) across all periods. This removes the endogenous response of IPO activity to fluctuations in the business cycle.¹⁸

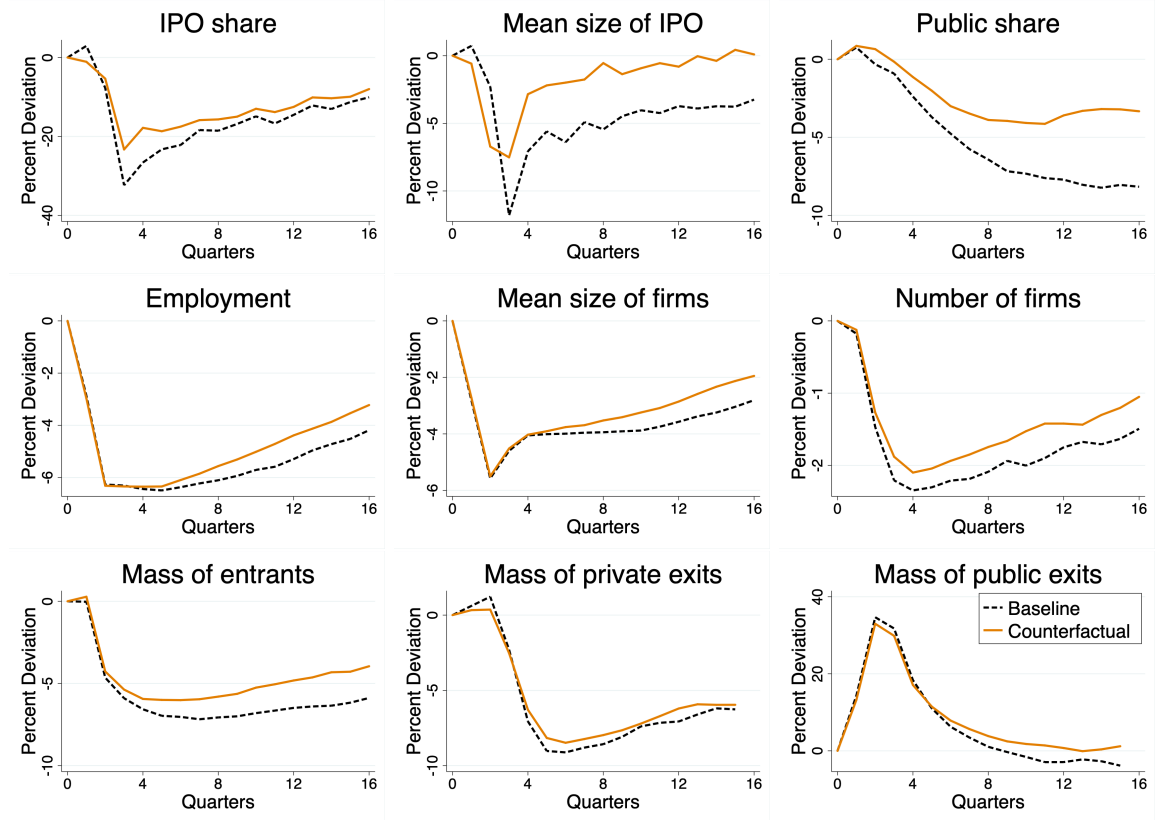
The first row of Figure 12 demonstrates that imposing a fixed IPO threshold damp-

¹⁸While IPO decisions in this setting are no longer the outcome of firms' optimization but are externally constrained, this approach isolates the contribution of IPO cyclicalities to aggregate employment dynamics.

ens the procyclical response of IPO activity. The IPO share—defined as the number of IPOs relative to the stock of private firms—remains relatively stable, and the decline in the average size of IPO firms is notably smaller than in the baseline. As the IPO share determines the entry margin of public firms (IPO), this stabilization leads to a relatively constant share of public firms, with minimal variation over time. The second and third rows of Figure 12 display the impulse responses of aggregate outcomes. While the initial decline in aggregate employment is comparable to that in the baseline scenario, the recovery proceeds more swiftly, avoiding the 2–3 quarter delay observed under endogenous IPO dynamics. This accelerated rebound is supported by improvements along both the intensive and extensive margins. On the intensive margin, although the average firm size contracts initially to a similar extent, it begins to recover earlier and more decisively. This faster recovery is primarily driven by the dynamics of public firms. Upon going public, firms experience a relaxation of financial constraints—both through the capital injection received at the IPO, which raises their net worth, and through enhanced access to external financing due to increased transparency and reduced information asymmetries. Since IPO firms in the model are predominantly small and productive—i.e., those previously financially constrained—they experience rapid post-IPO growth, contributing substantially to the rise in average firm size. Moreover, the stabilized IPO threshold in the counterfactual scenario leads to a larger share of public firms in the economy. This shift enhances allocative efficiency, as public firms, with fewer financing constraints, grow faster. As a result, the economy achieves a more efficient distribution of capital, reinforcing the recovery in aggregate productivity and employment through both direct firm-level dynamics and compositional changes.

On the extensive margin, the number of firms declines by approximately 2 percent in the first year following the shock—0.4 percentage points less than in the baseline. This more moderate decline is primarily driven by differences in entry dynamics. In the counterfactual scenario, the number of entrants falls by at most 5 percent, which is 2 percentage points less than in the baseline, and recovers more rapidly. However, this increase in the number of firms due to higher entry is partially offset by a rise in exit rates among both private and public firms. The underlying mechanism through which IPO cyclicalities affects entry and exit dynamics hinges on the predictability of capital access. A fixed IPO threshold ensures that firms meeting the criterion gain access to public equity markets regardless of aggregate conditions. For potential entrants, this predictability raises the expected future value of entry by guaranteeing access to

Figure 13: Response to a negative productivity shock – Counterfactual II



Notes. The black dashed line represents the response of the baseline model. The yellow solid line illustrates the response in the counterfactual scenario, in which IPO cyclicalities related to the capital injection is shut down.

external finance upon reaching the IPO threshold after the market entry. As a result, entry becomes more attractive even in downturns, and more firms choose to enter the market. However, some of these entrants may fall short of the productivity required for long-term viability. Initially motivated by the perceived opportunity to go public, such firms may struggle to survive in adverse conditions and exit prematurely. Thus, while entry increases, exit rates may also rise, reflecting the trade-off between broader access to finance and the potential for misallocation when selection is weakened.

In the second counterfactual scenario, I shut down the cyclicalities of capital injections by fixing them at their stationary level, allowing variation only through firm-specific factors such as idiosyncratic productivity and net worth. As a result, cyclical variation in IPO incentives arises solely from the variation in marginal cost of capital. In this scenario, firms make endogenous IPO decisions. As shown in the first row of

Figure 13, both the decline in the average size of IPO firms and the drop in the number of IPOs are substantially mitigated. Compared to the first counterfactual, the reduction in IPO activity is less pronounced, but its impact on the public firm share is not negligible. While the public firm share still declines following the negative shock, the contraction is much slower, with the gap relative to the baseline widening over time. This outcome reflects the dominance of the entry (IPO) margin: despite a slightly higher exit rate in the counterfactual, the elevated number of IPOs more than offsets exits, leading to a faster recovery in the public firm share.

Under this counterfactual scenario, employment initially declines by a similar magnitude as in the baseline but begins to recover more rapidly from the fourth quarter onward. This accelerated recovery is supported by both the intensive and extensive margins. On the intensive margin, the average firm size contracts following the shock, as in the baseline, but rebounds more quickly beginning in the fourth quarter. The increased injection leads more IPOs, who all become public firms. Then, the underlying mechanism mirrors that of the first counterfactual: as public firms reallocate resources more efficiently, the faster recovery in their share alleviates capital misallocation, thereby supporting the expansion of average firm size. The total number of firms declines by up to 2 percent—approximately 0.3 percentage points more than in the baseline. While this difference is more pronounced than in the first counterfactual, the underlying mechanism is attributed to entry and exit dynamics. A stable stock price allows potential entrants to anticipate more reliable IPO access during downturns, thereby raising the value of entry. As a result, the number of new entrants falls by 6 percent in the fourth quarter—0.6 percentage points more than in the baseline—with the gap widening over time. Compared to the earlier counterfactual, the change in entry is less substantial; however, the larger decline in the total number of firms is primarily driven by exit dynamics. Unlike the first counterfactual, where exogenous IPO access allowed low-productivity firms to enter and subsequently exit, the endogenous IPO threshold in this setting screens out such firms, preventing an increase in exits. At the same time, stable stock prices improve capital access for public firms, enabling them to remain operational. Thus, while both entry and exit decline, the reduction in exits is less severe, moderating the overall firm decline.

Table 7 summarizes the quantitative effects of the baseline and counterfactual scenarios in the fourth quarter following a negative aggregate productivity shock. In the baseline, employment declines by 6.44 percent. In the first counterfactual, where

Table 7: Quantitative responses: baseline vs. counterfactuals

	Baseline	Counter. I	Dev. (%)	Counter. II	Dev. (%)
Employment	-6.44	-5.77	10.40	-6.35	1.40
Mean size of firms	-4.05	-3.68	9.14	-4.03	0.49
Number of firms	-2.35	-1.99	15.32	-2.10	10.64
Public firm share	-2.39	1.12	146.86	-1.14	52.30
Entry rate	-6.57	-5.44	17.20	-5.94	9.59

Notes. The table presents the quantitative responses to a 1 percent negative aggregate productivity shock in the fourth quarter, comparing the baseline scenario with two counterfactual scenarios: (I) IPO cyclicalities—including both selection effects and the number of IPOs—is eliminated; (II) Procyclicality of capital injection is removed.

IPO access is exogenously stabilized, the decline is mitigated to 5.77 percent—a 10.40 percent smaller drop. This effect is driven by a 9.14 percent smaller contraction in average firm size and a 15.32 percent smaller reduction in the number of firms. The second counterfactual, which shuts down the cyclicalities of capital injection, shows a more modest attenuation: employment falls by 6.35 percent, corresponding to a 1.4 percent improvement relative to the baseline. This effect is primarily driven by a 10.64 percent smaller reduction in the number of firms, while the difference in average firm size is minimal (0.49 percent).

5 Conclusion

This paper examines the contribution of IPOs on aggregate employment dynamics, building on the well-established observation that IPO activity is procyclical. Empirical evidence shows that not only is the number of IPOs procyclical, but there is also a selection effect in terms of firm size and differential post-IPO employment growth patterns that vary according to IPO timing within the business cycle. Specifically, firms going public during downturns experience employment growth that is at least 10 percentage points slower over several years post-IPO compared to firms going public during expansions. This growth disparity is associated with the smaller capital raised at IPO during contraction periods.

To investigate the broader implications of this pattern, I develop a heterogeneous firm dynamics model in which private firms decide on going public, while both private and public firms make decisions regarding exit, investment, and employment under borrowing constraints. The model incorporates a procyclical stochastic discount fac-

tor, capturing the observed procyclicality in IPO numbers, capital raised at IPOs, and selection patterns documented in the data. In counterfactual analyses where I shut down the cyclicity in IPO composition and frequency, the results indicate that IPO cyclicity amplifies the impact of aggregate productivity shocks on employment. In particular, without IPO cyclicity, aggregate employment would decrease by 5.8 percent, compared to a 6.4 percent decrease in the baseline model. This amplification operates through two primary channels. First, a lower share of public firms during downturns exacerbates capital misallocation, as public firms typically operate more efficiently due to relaxed borrowing constraints. Second, the lower propensity for firms to go public in recessions diminishes the value of private firms, reducing the incentive for new entrants and thereby slowing recovery on the extensive margin. An alternative counterfactual analysis, in which the cyclicity of stock prices is eliminated, shows consistent results with regard to the underlying mechanism.

These findings carry important policy implications. Encouraging more firms to go public during or shortly after recessions could stabilize the labor market and accelerate recovery. Policies that reduce IPO fixed costs (as with the JOBS Act), provide incentives for public listing during downturns, or mitigate stock market volatility could counteract IPO cyclicity, maintaining a higher share of public firms, improving capital allocation efficiency, and encouraging new market entries.

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Appendix

A Empirical Evidence

A.1 Cyclical Number of IPO

Table 8: Correlation Matrix of Number of IPO and Business Cycle Measures

	Number of IPOs	PE Ratio	GDP
<i>Panel A. All (1980Q1-2021Q4)</i>			
Number of IPOs	1.00		
PE Ratio (detrended)	0.27***	1.00	
GDP (logged, detrended)	0.04	0.50***	1.00
<i>Panel B. Post-2000Q1</i>			
Number of IPOs	1.00		
PE Ratio (detrended)	0.66***	1.00	
GDP (logged, detrended)	0.23**	0.63***	1.00

Figure 1 reveals fluctuations in other periods, which are likely to be related to the stock market performance (Tran and Jeon, 2011). Panel A of Table 8 provides a correlation analysis between IPO numbers and the cyclical components of output and stock market performance, represented by detrended log real GDP and the detrended Shiller price-to-earnings ratio. The results show a stronger correlation with the stock market's cyclical component (0.27) than with GDP (0.04), highlighting the significant impact of stock market conditions on IPO activities beyond recessionary periods. However, a structural shift around 2000 marked by a significant drop in IPO frequency and its variance alters this pattern. Post-2000 correlation analysis in Panel B reveals increased correlations between both stock market performance and GDP with the number of IPOs, with a notable rise in the correlation with GDP. This shift indicates not only a reduction in the number of IPOs but also an intensified cyclical pattern in the IPO market since 2000.

Table 9: Descriptive statistics of IPO firms

	N	Mean	Std. Dev.	p10	p25	p50	p75	p90
<i>A. Expansion</i>								
Employment (1K)	9,815	1.48	7.39	0.02	0.05	0.18	0.72	2.67
Asset (1M)	10,256	30,482	151,858	94.47	413.57	2,207	11,364	48,482
Sales (1M)	10,191	21,216	110,765	0.00	95.40	1,105	6,883	32,338
Age	4,346	14.69	19.49	2.00	4.00	8.00	16.00	34.00
Book value per share	10,225	184.61	761.56	2.17	35.37	118.82	284.23	670.80
Debt-to-asset ratio	9,565	0.52	0.75	0.08	0.17	0.35	0.63	1.02
Sale of stock (1M)	10,036	4,030	8,113	0.00	89.14	622.60	3,629	12,001
Exit age	7,814	7.88	6.43	2.00	3.00	6.00	11.00	17.00
<i>B. Contraction</i>								
Employment (1K)	609	2.18	14.35	0.01	0.03	0.12	0.62	2.70
Asset (1M)	651	48,206	249,736	44.02	85.99	369.40	4,018	45,856
Sales (1M)	649	31,711	187,398	2.37	35.14	246.81	1,788	24,247
Age	90	16.04	22.40	3.00	5.00	10.00	16.00	27.50
Book value per share	645	218.38	537.96	1.37	10.81	47.45	166.45	687.19
Debt-to-asset ratio	637	0.55	0.72	0.09	0.20	0.41	0.67	0.95
Sale of stock (1M)	642	3,455	9,564	0.00	3.76	54.41	338.89	10,444
Exit age	521	8.53	7.51	1.00	3.00	6.00	12.00	19.00

Notes. This table presents the summary statistics in the year of IPO for firms categorized as expansionary or contractionary based on the timing of the IPO. Employment is measured in thousands. Assets, sales, and stock sales are measured in millions of U.S. dollars, while book value per share is measured in U.S. dollars. All figures are adjusted to 2012 real terms. Age information, based on the year of foundation, is sourced from FactSet and the Field-Ritter dataset of company founding dates, covering 40 percent of IPO firms after excluding those with negative ages and discrepancies of more than 20 years between the two sources. The remaining data are sourced from Compustat/CRSP.

A.2 Descriptive Statistics

Table 9 presents the descriptive statistics in the year of IPO for the two IPO cohorts.¹⁹ First, during contraction periods, the employment distribution of IPO firms exhibits greater variance. Although the average size for the contraction cohort is larger, this is likely driven by a few firms above the 90th percentile, as the average size up to the 75th percentile is consistently smaller compared to the expansion cohort. Similar patterns are observed in other dimensions of size, such as assets and sales, as well as the age, book value per share, and sale of common and preferred stock indicating the amount of capital injection raised at IPO. This suggests that, excluding the firms at the extreme right of the distribution, the contraction cohort is generally smaller and younger and

¹⁹Considering the possibility of a time lag, I also take the average up to a year after the IPO as a robustness check, which yields consistent results.

raises less amount of capital compared to the expansion cohort.

Table 10: Exit rates by IPO cohort

	Expansion	Contraction
Exit rate	0.564	0.565
Merger and acquisition	0.239	0.198
Bankruptcy or unknown	0.205	0.292
Delist (Back to private)	0.083	0.044

Table 10 presents the exit rates for the IPO cohorts. The exit rate is defined as the likelihood of being delisted from the Compustat/CRSP dataset within 10 years. The data indicate that both cohorts exhibit a consistent exit rate of 56 percent within this period. However, the types of exits differ between the cohorts. For the expansion cohort, mergers and acquisitions constitute the majority of exits, whereas for the contraction cohort, closures or bankruptcies are more prevalent. The incidence of firms returning to private status is the least common for both cohorts but occurs at twice the rate in the expansion cohort compared to the contraction cohort. The five-year exit rates are 35 percent for both cohorts, with a consistent composition.

A.3 Procyclical Capital Injection

I run a cross-sectional Ordinary Least Squares (OLS) regression of capital injection—the sale of stock (*sstk* in Compustat) during the IPO period—on the cyclical variable at the time of the IPO.²⁰

$$\log(sstk_{it}) = \beta Cycle_{it} + \Gamma \mathbf{X}_{it} + \epsilon_{it} \quad (23)$$

The coefficient of interest, β , captures the relationship between capital injection and the state of the economy at the time of the IPO, measured either by a dummy variable for the contraction cohort or by the detrended log of real GDP. The control variables, $\mathbf{X}_{it}$, include employment, sales per worker, capital intensity, and debt-to-asset ratio in the year of the IPO, as well as industry and state fixed effects.²¹

²⁰A robustness check is conducted using the aggregate sale of stocks in the first three years post-IPO to account for potential time lags.

²¹For capital intensity and the debt-to-asset ratio, I specifically use data of the period only prior to the IPO, as the sale of stock is significantly collinear with these variables post-IPO.

Table 11: Capital injection at IPO over the business cycle

	(1)	(2)	(3)
Log sale of stock at IPO			
IPO in Contraction	-0.378* (-2.22)		
Detrended Log GDP		0.295*** (13.87)	0.195*** (7.50)
Detrended PE Ratio			0.062*** (6.67)
Firm Characteristics at IPO	✓	✓	✓
Industry FE	✓	✓	✓
State FE	✓	✓	✓
Observations	5,096	5,096	5,096
R^2	0.297	0.322	0.328

Notes. t -statistics in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The table presents regression results of stock sales at IPO on the business cycle index using cross-sectional data of IPO firms. The cyclical components of log GDP and Shiller PE ratio are detrended using HP filter. 'Firm Characteristics at IPO' includes employment, sales per worker, capital intensity, and debt-to-asset ratio. For capital intensity and debt-to-asset ratio, data prior to IPO capital internalization are used. Firms exiting within two years post-IPO are excluded.

In Table 11, Column (1) shows that firms in the contraction cohort raise 38 percent less capital at IPO compared to those in the expansion cohort, holding other characteristics constant. To be specific, column (2) indicates that a 1 percentage point increase in GDP at the time of IPO results in a 30 percent greater amount of capital injection. Column (3) additionally considers the cyclicalities of stock market performance represented by Shiller price-to-earnings (PE) ratio to confirm that the relationship with economic output cyclicalities is not simply due to stock market fluctuations. This confirms that the amount of capital injection is indeed related to the economic state at the time of the IPO, consistent with existing literature (Tran and Jeon, 2011).

B Numerical Appendix

B.1 Interest rate

The interest rate in the model is defined as the inverse of the expected stochastic discount factor. By using the law of motion of aggregate productivity shock in (3), the one-period interest rate is expressed as

$$\begin{aligned}
\frac{1}{R_t} &= E[M(z_t, z_{t+1})] = E[\beta \exp(\phi_0 z_t + \phi_1 z_{t+1})] \\
&= \beta E[\exp(\phi_0 z_t + \phi_1 z_{t+1})] \\
&= \beta E[\exp(\phi_0 z_t + \phi_1 \rho_z z_t + \phi_1 \epsilon_{z,t})] \\
&= \beta \exp(\phi_0 z_t + \phi_1 \rho_z z_t) E[\exp(\phi_1 \epsilon_{z,t})] \\
&= \beta \exp(\phi_0 z_t + \phi_1 \rho_z z_t) \exp\left(0 + \frac{\phi_1^2 \sigma_z^2}{2}\right) \\
&= \beta \exp\left((\phi_0 + \phi_1 \rho_z) z_t + \frac{\phi_1^2 \sigma_z^2}{2}\right) \\
R_t &= \frac{1}{\beta} \exp\left(-(\phi_0 + \phi_1 \rho_z) z_t - \frac{\phi_1^2 \sigma_z^2}{2}\right).
\end{aligned}$$

B.2 Labor decision

Optimal level of labor is determined by maximizing the profit function, expressed as

$$l = \operatorname{argmax}_l sz(k^\alpha l^{1-\alpha})^\eta - wl$$

First order condition is

$$\begin{aligned}
(sz)\eta(k^\alpha l^{1-\alpha})^{\eta-1}(1-\alpha)k^\alpha l^{-\alpha} - w &= 0 \\
(sz)(1-\alpha)\eta k^\alpha l^{(1-\alpha)\eta-1} &= w \\
l^{(1-\alpha)\eta-1} &= \frac{w}{(1-\alpha)\eta(sz)} k^{-\alpha\eta}.
\end{aligned}$$

Therefore,

$$l^* = \left(\frac{w}{(1-\alpha)\eta(sz)} k^{-\alpha\eta}\right)^{\frac{1}{(1-\alpha)\eta-1}} = \left(\frac{(1-\alpha)\eta}{w}\right)^{\frac{1}{1-(1-\alpha)\eta}} (sz)^{\frac{1}{1-(1-\alpha)\eta}} k^{\frac{\alpha\eta}{1-(1-\alpha)\eta}}.$$

Substituting the optimal labor into the profit function yields

$$\begin{aligned}
\pi &= szk^{\alpha\eta}l^{(1-\alpha)\eta} - wl - (r + \delta)k + (1 + r)a \\
&= \left(\frac{(1 - \alpha)\eta}{w} \right)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} (sz)^{\frac{1}{1-(1-\alpha)\eta}} k^{\frac{\alpha\eta}{1-(1-\alpha)\eta}} \\
&\quad - w \left(\frac{(1 - \alpha)\eta}{w} \right)^{\frac{1}{1-(1-\alpha)\eta}} (sz)^{\frac{1}{1-(1-\alpha)\eta}} k^{\frac{\alpha\eta}{1-(1-\alpha)\eta}} - (r + \delta)k + (1 + r)a \\
&= \left(((1 - \alpha)\eta)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} - ((1 - \alpha)\eta)^{\frac{1}{1-(1-\alpha)\eta}} \right) w^{\frac{-(1-\alpha)\eta}{1-(1-\alpha)\eta}} (sz)^{\frac{1}{1-(1-\alpha)\eta}} k^{\frac{\alpha\eta}{1-(1-\alpha)\eta}} \\
&\quad - (r + \delta)k + (1 + r)a \\
&= (1 - (1 - \alpha)\eta) ((1 - \alpha)\eta)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} w^{-\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} (sz)^{\frac{1}{1-(1-\alpha)\eta}} k^{\frac{\alpha\eta}{1-(1-\alpha)\eta}} \\
&\quad - (r + \delta)k + (1 + r)a \\
&= (1 - (1 - \alpha)\eta) \left(\frac{(1 - \alpha)\eta}{w} \right)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} (sz)^{\frac{1}{1-(1-\alpha)\eta}} k^{\frac{\alpha\eta}{1-(1-\alpha)\eta}} - (r + \delta)k + (1 + r)a.
\end{aligned}$$

B.3 Capital allocation

Optimal level of capital investment is determined by maximizing the profit function of the next period, expressed as

$$k' = \operatorname{argmax} \mathbb{E}_{s', z'} [\pi(a', s', z')] \quad \text{s.t.} \quad k' \leq \frac{1}{1 - \theta} a'$$

The first order condition is

$$\begin{aligned}
&\frac{\partial}{\partial k'} \mathbb{E}_{s', z'} \left[(1 - (1 - \alpha)\eta) \left(\frac{(1 - \alpha)\eta}{w} \right)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} (s'z')^{\frac{1}{1-(1-\alpha)\eta}} k'^{\frac{\alpha\eta}{1-(1-\alpha)\eta}} - (r(z') + \delta)k' + (1 + r)a' \right] \\
&= \mathbb{E}_{s', z'} \left[\frac{\partial}{\partial k'} \left((1 - (1 - \alpha)\eta) \left(\frac{(1 - \alpha)\eta}{w} \right)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} (s'z')^{\frac{1}{1-(1-\alpha)\eta}} k'^{\frac{\alpha\eta}{1-(1-\alpha)\eta}} - (r(z') + \delta)k' + (1 + r)a' \right) \right] \\
&= \mathbb{E}_{s', z'} \left[\alpha\eta \left(\frac{(1 - \alpha)\eta}{w} \right)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} (s'z')^{\frac{1}{1-(1-\alpha)\eta}} k'^{\frac{\eta-1}{1-(1-\alpha)\eta}} - (r(z') + \delta) \right] = 0.
\end{aligned}$$

Rearranging them,

$$\begin{aligned} \alpha\eta((1-\alpha)\eta)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} k'^{\frac{\eta-1}{1-(1-\alpha)\eta}} \mathbb{E}_{s',z'} \left[\left(\frac{s'z'}{w^{(1-\alpha)\eta}} \right)^{\frac{1}{1-(1-\alpha)\eta}} \right] &= \mathbb{E}_{z'}[r(z') + \delta] \\ k'^{\frac{\eta-1}{1-(1-\alpha)\eta}} &= \frac{\mathbb{E}_{z'}[r(z') + \delta]}{\alpha\eta((1-\alpha)\eta)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} \mathbb{E}_{s',z'} \left[\left(\frac{s'z'}{w^{(1-\alpha)\eta}} \right)^{\frac{1}{1-(1-\alpha)\eta}} \right]} \\ k' &= \left(\frac{\alpha\eta((1-\alpha)\eta)^{\frac{(1-\alpha)\eta}{1-(1-\alpha)\eta}} \mathbb{E}_{s',z'} \left[\left(\frac{s'z'}{w^{(1-\alpha)\eta}} \right)^{\frac{1}{1-(1-\alpha)\eta}} \right]}{\mathbb{E}_{z'}[r(z') + \delta]} \right)^{\frac{1-(1-\alpha)\eta}{1-\eta}}. \end{aligned}$$

Therefore, the first-best capital investment is

$$k' = (\alpha\eta)^{\frac{1-(1-\alpha)\eta}{1-\eta}} ((1-\alpha)\eta)^{\frac{(1-\alpha)\eta}{1-\eta}} \left(\frac{\int \int \left(\frac{s'z'}{w^{(1-\alpha)\eta}} \right)^{\frac{1}{1-(1-\alpha)\eta}} dH(s'|s) dG(z'|z)}{\int (r(z') + \delta) dG(z'|z)} \right)^{\frac{1-(1-\alpha)\eta}{1-\eta}},$$

which implies the optimal capital choice given a' as

$$k'^* = \max \left(\frac{a'}{1-\theta}, (\alpha\eta)^{\frac{1-(1-\alpha)\eta}{1-\eta}} ((1-\alpha)\eta)^{\frac{(1-\alpha)\eta}{1-\eta}} \left(\frac{\int \int \left(\frac{s'z'}{w^{(1-\alpha)\eta}} \right)^{\frac{1}{1-(1-\alpha)\eta}} dH(s'|s) dG(z'|z)}{\int (r(z') + \delta) dG(z'|z)} \right)^{\frac{1-(1-\alpha)\eta}{1-\eta}} \right).$$

B.4 Price of Share

The price of share described in (11) can be reformulated as

$$\chi p(a_t, k_t, s_t, z_t) = \mathbb{E}_t \left[M(z_t, z_{t+1}) \max\{(\chi d_{t+1} + \chi p(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1})), \chi d_{t+1}^{x,pb}\} \right],$$

where $d_{t+1} = \pi_{t+1}(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1}) - f - a_{t+2}(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1})$ and π_{t+1} is as described in (13). By cancelling out χ , it is reduced to

$$p(a_t, k_t, s_t, z_t) = \mathbb{E}_t \left[M(z_t, z_{t+1}) \max\{(d_{t+1} + p(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1})), d_{t+1}^{x,pb}\} \right]. \quad (24)$$

In the meantime,

$$\begin{aligned}
& \mathbb{E}_t \left[M(z_t, z_{t+1}) V^{pb}(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1}) \right] \\
&= \mathbb{E}_t \left[M(z_t, z_{t+1}) \max\{\tilde{V}^{pb}(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1}), V^x(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1})\} \right] \\
&= \mathbb{E}_t \left[M(z_t, z_{t+1}) \max\{(1 - \chi)d_{t+1} + \mathbb{E}_{t+1} \left[M(z_{t+1}, z_{t+2}) V^{pb}(a_{t+2}, k_{t+2}, s_{t+2}, z_{t+2}) \right], (1 - \chi)d_{t+1}^{x,pb}\} \right] \\
&= (1 - \chi) \mathbb{E}_t \left[M(z_t, z_{t+1}) \max\{d_{t+1} + \frac{1}{1 - \chi} \mathbb{E}_{t+1} \left[M(z_{t+1}, z_{t+2}) V^{pb}(a_{t+2}, k_{t+2}, s_{t+2}, z_{t+2}) \right], d_{t+1}^{x,pb}\} \right]
\end{aligned}$$

which can be reformulated as

$$\begin{aligned}
& \frac{1}{1 - \chi} \mathbb{E}_t \left[M(z_t, z_{t+1}) V^{pb}(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1}) \right] \\
&= \mathbb{E}_t \left[M(z_t, z_{t+1}) \max\{d_{t+1} + \frac{1}{1 - \chi} \mathbb{E}_{t+1} \left[M(z_{t+1}, z_{t+2}) V^{pb}(a_{t+2}, k_{t+2}, s_{t+2}, z_{t+2}) \right], d_{t+1}^{x,pb}\} \right].
\end{aligned} \tag{25}$$

The condition $p(a_t, k_t, s_t, z_t) = \frac{1}{1 - \chi} \mathbb{E}_t \left[M(z_t, z_{t+1}) V^{pb}(a_{t+1}, k_{t+1}, s_{t+1}, z_{t+1}) \right]$ makes the Equation (24) and (25) equivalent. Therefore,

$$\chi p(a, s, z) = \frac{\chi}{1 - \chi} \mathbb{E}_t \left[M(z, z') V^{pb}(a', s', z') \right].$$